User Perception and Pain Points in IoT-enabled Smart Farming Systems

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***Abstract*—This study is an analytical survey aimed at evaluating user satisfaction, identifying issues related to sensor usage, and improving UX/UI for a better user experience. We will gather feedback from actual users to understand their experiences and the factors affecting sensor performance, particularly in terms of stability and user interface challenges. A questionnaire will be designed to survey IoT users across five locations in Chiang Rai province, including both new and existing users. By analyzing the technical and UX/UI challenges encountered, we aim to develop solutions that enhance sensor connectivity stability and align with users' needs. Ultimately, this study seeks to identify pain points in the use of sensors within IoT-enabled smart farming and address UI/UX challenges faced by users, improving usability and system interaction.**

***Index Terms*—User Perception, IoT, Smart Farming**

I. Introduction

In recent years, the Internet of Things (IoT) technology has been widely adopted in agriculture to promote smart farming practices. It plays a crucial role in supporting real-time environmental data monitoring, enhancing decision-making accuracy, and improving efficiency in the cultivation process [1]. However, despite the high potential of IoT technology in advancing the agricultural sector, empirical evidence from user experiences reveals several limitations, particularly among farmers who lack technological expertise [2].

This research is guided by three key questions aimed at addressing critical issues in the adoption and effectiveness of IoT-based smart farming systems. Firstly, it explores the problems and limitations that users encounter, particularly those related to the stability of sensor connectivity and the usability of system interfaces. Secondly, the study seeks to determine how solutions can be designed to improve sensor connectivity and enhance UX/UI design in order to optimize both performance and user experience. Lastly, it investigates to what extent an improved IoT smart farming system aligns with user needs, focusing on system stability, ease of use, and its effectiveness in solving practical agricultural challenges. These research questions serve as the foundation for understanding user pain points and for developing resilient, user-centered technologies that can support sustainable and efficient agricultural practices.

Motivated by these questions and the increasing need for resilient, user-centered agricultural technologies, this study seeks to bridge the gap between system performance and real-world usability. In particular, the focus is on enhancing sensor connectivity stability [12], simplifying user interfaces [11], and incorporating responsive alert mechanisms to improve both user trust and operational decision-making [13]. These improvements are especially relevant in rural regions such as Chiang Rai Province, where farmers may lack access to technical support but could benefit significantly from practical, well-designed digital tools.

The primary objective of this research is to explore and analyze the challenges faced by users in utilizing IoT-enabled smart farming systems, with particular emphasis on issues related to sensor connectivity instability in environments that experience highly variable conditions, such as adverse weather. Additionally, the research aims to identify pain points and understand the genuine needs of farmers through the analysis of data collected from questionnaires, providing insight into areas that require improvement. Based on these findings, the study seeks to develop solutions that enhance sensor connectivity stability, improve the accuracy and timeliness of alert mechanisms, and optimize user experience (UX) and user interface (UI) design to make the system more user-friendly and practical. Ultimately, the goal is to evaluate the improved IoT smart farming system to ensure that it meets user needs effectively in terms of stability, ease of use, and the ability to solve problems efficiently, thereby fostering long-term adoption and sustainable use in agricultural contexts.

II. Literature Review

Smart farming technology has transformed traditional agriculture into a data-driven and automated system. Due to the fluctuating impacts of climate conditions on farmers' livelihoods, the Internet of Things (IoT) has become increasingly essential in enabling real-time data management and the efficient use of resources [14]. Technological advancements in agricultural IoT have focused primarily on sensor systems [15], connectivity issues, user satisfaction, and UX/UI design within the smart farming framework [16].

The objective of this research is to identify gaps in IoT-based smart farming systems that still face challenges in practical implementation [17]. Previous studies have identified several technical barriers, including unstable sensor connectivity and the lack of human-centered design principles, which contribute to usability difficulties and hinder user adoption. This study aims to analyze common problems encountered in IoT-based agricultural systems, ranging from technical failures to usability issues, in order to evaluate whether these issues have been resolved or remain persistent. Gaining this understanding will support the development of improved systems capable of meeting the diverse needs of users in the agricultural sector [18].

This paper provides an overview of smart farming concepts and technologies related to IoT, such as sensor networks, cloud-based data management, and communication protocols. It then explores various approaches and case studies of IoT adoption, with a focus on system reliability and user interface design. Finally, the study highlights how it addresses existing gaps in the current body of knowledge, particularly in enhancing IoT system connectivity and improving UX/UI for agricultural users throughout the research process.

A. Context

Smart farming, driven by Internet of Things (IoT) technologies, is transforming agriculture by improving efficiency, productivity, and sustainability [19]. These technologies allow farmers to monitor and manage key factors such as soil moisture, temperature, humidity, and pest presence using real-time data from sensors [20]. In countries like Thailand, IoT systems such as automated irrigation and environmental monitoring have been gradually introduced.

Smart farming integrates IoT systems to enable precision agriculture, offering automation, data-driven decision-making, and remote control. These systems typically consist of environmental sensors, cloud platforms, wireless communication modules, and mobile applications [21]. Successful implementations have demonstrated improvements in water efficiency, fertilizer use, and crop yields through smart irrigation, weather monitoring, and pest detection [22], [23]. However, deployment in resource-limited settings often requires customized solutions that balance cost, robustness, and scalability.

However, most studies emphasize technical optimization and system performance, while farmer perspectives, system usability, and environmental adaptability receive limited attention. Factors such as sensor instability under extreme weather e.g., storms, dust, and heat remain barriers to adoption [24],[25]. If the system fails or data is inaccurate, it may disrupt farming operations, resulting in reduced trust and adoption.

In rural contexts, technology adoption often faces challenges such as low digital literacy, limited internet coverage, and inadequate infrastructure. Studies have shown that while the perceived benefits of IoT are strong, usability issues including complex interfaces, technical jargon, and maintenance concerns limit widespread acceptance [26], [27]. Moreover, older farmers and those without prior technology experience may struggle to interpret sensor data or operate mobile applications, necessitating inclusive design strategies and local language support.

Moreover, real-time decision-making depends on reliable data. Sensors must function consistently, transmitting information through stable connections to cloud-based platforms [28]. These systems not only detect early signs of disease and environmental changes but also reduce human intervention, marking a shift toward automation in farming [29],[30].

While the technological benefits are clear, widespread and sustainable adoption requires understanding user experiences, perceived risks, cost concerns, and the broader human-technology interaction [31].

B. Trust Theory

The Technology Acceptance Model (TAM), developed by Davis (1989), identifies perceived usefulness and perceived ease of use as the two primary factors influencing technology adoption [32]. In the context of agriculture, technologies that are both beneficial and easy to understand stand a higher chance of being adopted, particularly in rural settings where farmers often face constraints in terms of time, resources, or technical expertise.

Building upon TAM, the Unified Theory of Acceptance and Use of Technology (UTAUT) by Venkatesh et al. (2003) incorporates four core elements that further explain user adoption behavior: performance expectancy (the belief that technology enhances outcomes), effort expectancy (the ease of using the technology), social influence (peer and community pressure or encouragement), and facilitating conditions (the availability of support infrastructure and resources) [33],[34]. These dimensions are especially relevant in agriculture, where the successful use of IoT systems often depends not only on the technology itself but also on external support and community acceptance.

Trust theory also plays a significant role in technology adoption, particularly in environments where data reliability and system performance are critical. Trust is formed when systems perform consistently, with low risk and high reliability [35]. In the case of smart farming, where faulty data can result in crop loss or misinformed decisions, trust in the IoT system encourages long-term use and reduces resistance to technology [36],[37],[38].

The Innovation Diffusion Theory (IDT), proposed by Rogers, highlights that adoption is influenced by factors such as relative advantage (perceived improvement over current practices), compatibility (fit with existing farming methods), and the ability to trial or observe the technology before full implementation [39],[40]. When farmers clearly see how a new system aligns with their needs and yields visible benefits like increased productivity or reduced input costs they are more likely to embrace the innovation.

The Information Quality Model emphasizes that high-quality data, accurate, consistent, and relevant is vital for effective decision-making. In smart agriculture, where decisions often depend on real-time sensor data (e.g., soil moisture or weather conditions), poor data quality can lead to inefficient resource use or crop damage. Ensuring that IoT systems produce reliable information is therefore fundamental to their value [48],[49].

Another relevant framework is Robustness Theory, which focuses on the ability of systems to maintain function and recover quickly in uncertain or changing environments. In agricultural contexts, where external conditions such as extreme weather or equipment failures are common, robust systems with backup and recovery mechanisms are essential. This includes quick data responses, system adaptability, and consistent operation under stress [10],[50],[51],[45].

Finally, the Cost-Benefit Theory explains adoption behavior based on an economic rationale: users evaluate whether the benefits of a new technology outweigh its costs. For farmers, this includes comparing installation, maintenance, and training costs against gains in productivity, labor savings, and resource efficiency [46],[47]. When the perceived or actual benefits surpass the costs, the likelihood of adoption increases closely linking this model with the concept of relative advantage in IDT [40].

C. Existing Work

The integration of IoT technologies in agriculture, often termed smart farming, has gained substantial traction for its promise to increase productivity, enable precision monitoring, and reduce resource waste. Prior research has explored various architectures and sensor-based systems, with many emphasizing real-time data collection, remote control, and environmental condition tracking. Tzounis et al. [48] outlined foundational IoT architectures using sensors for monitoring soil, temperature, humidity, and weather conditions. Similarly, Farooq et al. [49] emphasized the role of wireless communication and cloud platforms in offering visibility into farm conditions and enabling automated responses such as irrigation control. However, limitations remain. Gundes [50] discussed the issue of sensor stability, especially under harsh environmental conditions such as storms or power interruptions. Khedekar [51] further noted that while modern sensor systems have improved in precision, they still suffer from instability and inaccuracies in rural deployment scenarios.

On the more advanced end, Chen et al. [45] proposed an AIoT-based smart agriculture platform for pest detection. The system collected environmental data via wireless sensors and analyzed it using cloud-based big data analytics. Although the system demonstrated accuracy, its reliance on high computational power and stable internet connectivity posed practical challenges for remote farming areas. User experience is another important concern. Page et al. [46] examined farmers' perceptions of IoT technologies and found a gap in understanding, especially among users who preferred systems they could directly control. Usability and interface complexity were identified as significant barriers, particularly for non-technical users.

In terms of theoretical contributions, Kumar et al. [54] conducted a comprehensive review on event and anomaly detection in sensor-based IoT systems. However, their findings were largely theoretical, with limited real-world validation, especially in small-scale or resource-constrained farms. Regarding adoption behavior, Venkatesh et al. [55] and Davis [58] contributed insights through the UTAUT and TAM models, identifying key predictors such as performance expectancy and effort expectancy. Jayashankar et al. [56] and Naseer et al. [57] emphasized the role of trust in driving long-term IoT adoption, highlighting how consistent performance, data reliability, and perceived value influence users' willingness to adopt smart farming technologies.

Real-world implementation scenarios underscore the ongoing challenges associated with sensor reliability. Nattaphon Charoensiri et al. [59] developed a temperature and humidity alert system for electric control rooms using Arduino and LAN communication. Although the system aimed to enhance stability, it suffered from inconsistent data transmission and database connectivity issues, with LAN wiring introducing extra cost and complexity. Similarly, researchers at King Mongkut's University of Technology North Bangkok [60] designed an environmental monitoring system using a DHT21 sensor. While offering acceptable accuracy (±0.5°C for temperature and ±3% RH for humidity), the sensor's performance degraded under rapidly changing conditions or high humidity, raising concerns over its long-term reliability.

In another case study conducted in a mushroom farm in Nakhon Chai Si District, Nakhon Pathom, a team implemented a smartphone-controlled microclimate system using DHT22 sensors and Arduino [61]. The study revealed issues in both system responsiveness and sensor performance particularly in high-humidity environments, where the sensors often yielded unstable readings and delays in mobile app control were noticeable. These findings indicate that while IoT-based temperature and humidity monitoring systems are increasingly deployed in smart farming and environmental control, sensor reliability especially in harsh or variable conditions remains a key limitation. Selecting appropriate sensor models, managing signal stability, and considering real deployment conditions are essential for building dependable systems.

From the review of existing literature, it is evident that although numerous IoT-based solutions have been developed for smart farming focusing on environmental sensing, data-driven decision-making, and automation many challenges still persist. These include limitations in sensor stability under harsh conditions [43], delayed or inaccurate data transmission [52], lack of real-time alerts, and the physical durability of sensing systems in agricultural environments [60], [61]. Several studies reported inconsistent sensor performance, especially regarding temperature and humidity monitoring in high-humidity or rapidly fluctuating conditions [43], [61]. Furthermore, issues with interface usability, delayed system responsiveness, and infrastructure-related costs (e.g., wired connectivity) have also been frequently noted [59], [60]. These real-world limitations significantly hinder the reliability and long-term adoption of IoT systems among end-users, particularly non-technical farmers in remote areas [53].

Based on the aforementioned challenges, this study focuses on addressing three critical aspects: sensor stability, real-time alert mechanisms, and durability. Emphasis is placed on ensuring continuous environmental monitoring, minimizing data transmission delays through the use of appropriate communication protocols, and enhancing the resilience of hardware to match the demands of real agricultural environments. These improvements aim to increase system performance and reliability for long-term deployment in the field.

D. Conclusion

The review of existing literature reveals that while IoT-based Smart Farming technologies have made significant strides in enhancing agricultural efficiency, numerous practical challenges remain. Key barriers to widespread adoption include system instability under harsh environmental conditions, limited real-time reliability, and insufficient attention to user-centered design particularly for non-technical farmers in rural areas such as Chiang Rai.

Theoretical frameworks such as the Technology Acceptance Model (TAM), Unified Theory of Acceptance and Use of Technology (UTAUT), Innovation Diffusion Theory (IDT), and Trust Theory highlight the importance of usability, perceived value, and system robustness in driving adoption. Additionally, the importance of high-quality information and cost-benefit considerations underscores the necessity for systems that not only perform technically but also meet economic and experiential expectations of end-users.

Despite advancements in sensor design, cloud-based infrastructure, and AI-powered analytics, most existing systems still face constraints in offline capability, UX/UI adaptability, and real-world scalability. Therefore, this study addresses these gaps by developing a farmer-informed IoT system with hybrid anomaly detection, offline data resilience, and intuitive interface design tailored to the unique needs of remote agricultural communities.

III. METHODOLOGY

This study employs a design-driven research approach combined with Human-Centered Design (HCD) principles to improve IoT-based smart farming systems. The methodology focuses on addressing both technical issues and usability challenges by collecting user feedback through online surveys and in-depth interviews with farmers in Chiang Rai Province.

This chapter outlines the five key stages of the design process: identifying user needs, defining system requirements, designing the solution, developing the application, and testing the prototype. It also describes the methods used for both quantitative and qualitative data analysis to support the development of practical, user-focused system improvements.

A. Design Solution

This study aims to develop and enhance smart farming systems using IoT by focusing on both technological aspects (such as sensor connectivity stability and data accuracy) and user experience dimensions (such as ease of use and farmers' understanding). A Design-Driven Research approach is employed, integrating creative design methods with empirical evidence gathered from real users to develop solutions that address the deeper needs of farmers in the actual study area.

In the system design process, Human Centered Design (HCD) principles serve as the core methodology to ensure a user-focused development. Data was collected directly from local farmers through surveys, interviews, and usability testing. This approach enabled the research team to truly capture users' needs, pain points, and expectations, leading to a system design that genuinely reflects the realities of end users. The design solution is structured into 5 key stages:

- **Identifying User Needs** Identifying user needs is a critical step in addressing challenges associated with smart farming systems that utilize Internet of Things (IoT) technologies. This study aims to explore the problems and requirements of farmers, with a particular focus on the instability of sensor connectivity. The research targets two distinct groups of farmers: (1) those who are currently using IoT technology in their farming operations, and (2) those who previously adopted IoT systems but have since discontinued their use. The objective is to gather comprehensive insights into their experiences, expectations, and barriers to adoption, using both quantitative and qualitative research methods, including online questionnaires and on-site field studies.

  Initially, an exploratory study was conducted with six farmers in Chiang Rai province who are currently using IoT-based smart farming systems. A mixed-method approach was employed, combining an online questionnaire designed based on the Unified Theory of Acceptance and Use of Technology (UTAUT) with in-depth interviews. This approach enabled the collection of both quantitative and qualitative data on real user experiences, challenges, limitations, and expectations in rural contexts.

  The questionnaire consisted of 24 questions categorized into eight sections as follows

Section 1: General Information The first section collected general demographic and background information, such as the respondent's age and their experience with IoT technologies.
Section 2: System Reliability The second section examined system reliability by asking participants how often system failures occurred, the average weekly downtime, the level of user trust in the system, and how failure frequency affected confidence levels.
Section 3: Ease of Use The third section focused on ease of use, exploring the interpretability of IoT-generated data, the impact on agricultural productivity, and the extent of training or support required.
Section 4: Data Accuracy and Consistency The fourth section assessed the accuracy and consistency of the data, including the perceived precision of system recommendations, frequency of discrepancies between expected and actual outcomes, and the influence of environmental conditions on accuracy.
Section 5: Environmental Responsiveness Section five examined the system's responsiveness to environmental factors, such as its adaptability to weather changes, energy stability, and the impact of dust or pests on sensor functionality. This section also explored how these environmental variables influenced farmers' decision-making.
Section 6: Maintenance and Support The sixth section dealt with maintenance and technical support, investigating the frequency of system upkeep, ease of troubleshooting, and the need for recalibration.
Section 7: Cost and Benefit Perception Section seven addressed the perceived cost-benefit ratio of using IoT systems, including whether the systems contributed to higher productivity or reduced costs, and whether outcomes aligned with user expectations.
Section 8: Open-Ended Questions The final section comprised open-ended questions that invited participants to share their experiences and offer suggestions for future system development.

The findings revealed recurring issues faced by users, notably the instability of sensors particularly during heavy rain or flooding which leads to data inaccuracies or inconsistencies. In addition, limited internet infrastructure in certain areas causes delays in data transmission, impairing the effectiveness of real-time alerts. Farmers without a technical background also reported that the systems were overly complex and difficult to operate, leading to a lack of confidence in managing the technology independently.

At the same time, users clearly expressed the need for systems that are easy to operate, capable of displaying historical data, and equipped with timely and understandable alert functions. Such improvements would enable users to make prompt and informed decisions. These insights serve as a critical foundation for the design of a prototype system that directly addresses these real-world issues and aligns with the practical needs of farmers in rural areas. Based on the findings, the key user needs can be summarized into five main areas

- **Stable and accurate systems** that are resilient to extreme environmental conditions such as heavy rainfall and flooding.
- **Fast and continuous internet connectivity** to ensure real-time alert functionality.
- **User-friendly interfaces** suitable for individuals with limited technical knowledge.
- **Clear and timely alert systems** delivered via accessible channels such as LINE or SMS.
- **Historical data access** to support better decision-making and planning.

- **Defining System Requirements** To ensure that the IoT-based smart farming system effectively supports the agricultural practices and constraints of farmers in Chiang Rai Province, the system requirements were developed through a dual approach. This included empirical data from field research interviews, surveys, and observations and integration of conceptual frameworks such as TAM, UTAUT, Trust Theory, and Robustness Theory. The combination of these perspectives allowed the research team to design a system that is not only technically sound but also aligned with the practical needs and limitations of end users.
  - **Deriving Requirements from Field Data and Theory** The research began with gathering and analyzing responses from rural farmers, focusing on how they currently interact with technology and what challenges they face. From this process, two categories of requirements were clearly distinguished:
    1.Functional Requirements
    refer to the core operations the system must perform, such as capturing real-time sensor data, generating alerts, and displaying historical trends in a user-accessible way.
    2.Non-Functional Requirements
    deal with qualities that influence system performance and usability, including reliability, fault tolerance, ease of use, and adaptability across different farm settings.
  - **Organizing Requirements into Four Thematic Areas** To present the requirements in a form that reflects both theory and user context, they were grouped into four thematic clusters
  - **System Reliability and Robustness** Farmers operate in environments where electricity and internet service are not always stable. Therefore, the system must remain functional even during disruptions. It must store data locally when connections drop and synchronize automatically once conditions return to normal. The system must also prevent data loss, duplication, or corruption by employing robust fault-tolerance mechanisms. Scalability is another key concern: the system must be able to expand across

multiple farms and accommodate new sensor types as needed such as soil moisture, light intensity, or pH sensors.

  - **User-Centric Interface Design** Considering that many farmers in the target region have limited digital experience, the mobile application must be visually intuitive. This includes using large, clearly labeled icons, high-contrast colors, and Thai-language support. The interface should follow a minimalist design to reduce cognitive load and offer offline functionality. Navigation must be simple, with clear pathways for core tasks and features. Additionally, the system should include a feedback mechanism so that users can report problems or suggest improvements, ensuring ongoing refinement based on real user experience.
  - **Real-Time Monitoring and Alerts** The ability to monitor environmental data (like temperature and humidity) in real time is at the heart of the system. Data collection must continue uninterrupted even in remote locations. When sensor readings deviate from predefined thresholds, users should be alerted immediately via the mobile app. These alerts must be visually distinctive using colors, icons, and concise messages to ensure fast recognition. To maintain effectiveness, the system should deliver new data and notifications with a delay of no more than three seconds under standard network conditions.
  - **Data Accessibility and Historical Analysis** Farmers require access to past data in order to plan irrigation schedules, apply fertilizers efficiently, and evaluate crop outcomes. The system must present historical information in visual formats such as line graphs, trend charts, or data summaries. These tools must allow users to filter, zoom, or search for specific time frames. Moreover, the system should enable exporting data for printing or sharing, thus supporting planning and reporting activities in both digital and non-digital formats.

- **Designing a User-Oriented System for Farmers** Centered Design (UCD) is a crucial process that ensures technological systems effectively meet the real needs and expectations of end users. In this case, designing a system for farmers utilizing Internet of Things (IoT) technology requires a thorough understanding of the working conditions of farmers in rural areas, who often face limitations in both internet access and technical knowledge.

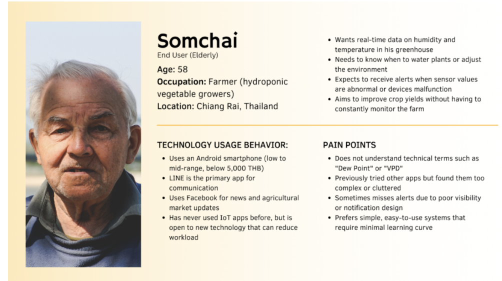

Fig. 1. DA1-User Persona

Figure DA1: User Persona for IoT system design, This persona, named Somchai, represents the typical end-user—a farmer in a rural area—detailing his demographic background, technology usage, specific needs, and pain points to guide the development of a user-centered, accessible, and high-accuracy IoT system.

Findings from user surveys indicate that farmers prefer systems that are easy to use and capable of delivering accurate information at the right time. Therefore, the system design must fully respond to these needs. Additional survey results reveal that farmers require a system with high stability and accuracy, especially during adverse weather conditions such as heavy rain or flooding, which can significantly impact sensor performance and data reliability.

To ensure effective operation, the system must support fast and stable internet connectivity, enabling real-time alerts for critical events, such as sensor malfunctions or significant environmental changes like temperature and humidity fluctuations that could affect crop growth. A system designed to address these specific needs will empower farmers to leverage IoT technology more effectively and contribute to sustainable agricultural productivity.

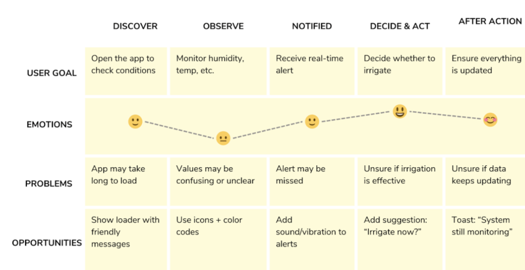

Fig. 2. DA2-User Journey Map.

Figure DA2: User Journey Map for the smart farming application., The map illustrates the farmer's experience across five stages (Discover to After Action), highlighting the user's Goals, Emotions, Problems encountered at each stage, and corresponding Opportunities for system improvement, emphasizing the need for clarity and timely alerts.

The design approach adheres to principles of simplicity and accessibility, ensuring usability for all, including farmers with limited technical backgrounds. Given the constraints of internet access and electronic devices in rural areas, the system was developed using frameworks grounded in Human-Centered Design (HCD) and the Unified Theory of Acceptance and Use of Technology (UTAUT). These frameworks help inform the development of practical functions that align with the actual usage context of farmers.

Moreover, the system must remain operational under challenging environmental conditions, such as heavy rain or intermittent internet connectivity. One key survey insight highlights that users demand a simple and intuitive system that can be operated without prior technical knowledge. Consequently, the user interface (UI) design emphasizes clarity using intuitive icons, distinct color schemes, and easy-to-understand Thai language to ensure usability without unnecessary complexity.

The system should also visualize historical data through graphs or charts, allowing users to review and make informed decisions in a timely manner. Furthermore, it must operate in both online and offline modes. In areas with poor internet connectivity, the system will locally store sensor data and synchronize it once a connection is restored. It must also be able to accurately monitor sensor status and notify users promptly when sensors malfunction.

For the development of the prototype application, tools like React-native were selected, as it supports both iOS and Android platforms. The application includes essential features such as historical data visualization and real-time notifications. It is also designed to remain functional in areas with unstable internet, ensuring reliable usability under all conditions.

To evaluate the effectiveness of the system, the research team has planned usability testing with a target group of farmers. This process will help determine whether the prototype meets user needs and expectations.

In summary, designing a user-centered system for farmers in rural areas requires careful consideration of various limitations, including low technological literacy, unstable internet connectivity, and environmental challenges that may affect sensor operations.A system that can promptly and reliably respond to these conditions will enhance agricultural efficiency and promote long-term productivity.

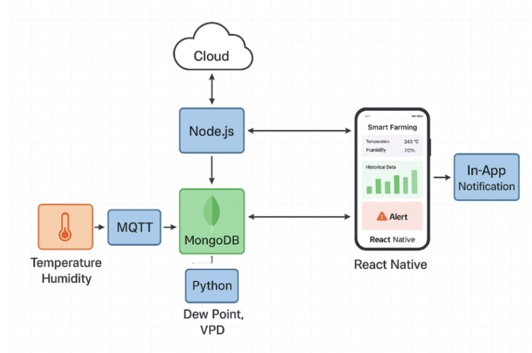

Fig. 3. DA3-System diagram

Figure DA2: User Journey Map for the smart farming application., The map illustrates the farmer's experience across five stages (Discover to After Action), highlighting the user's Goals, Emotions, Problems encountered at each stage, and corresponding Opportunities for system improvement, emphasizing the need for clarity and timely alerts.

- **System development** The development of the IoT-based smart farming system was structured into three main components: backend infrastructure, data simulation services, and the mobile application. The goal was to deliver a responsive, real-time system capable of monitoring environmental parameters, notifying users of anomalies, and providing an intuitive interface for non-technical users.
  - **Development Environment and Tools** The system was developed using Visual Studio Code (VS Code) as the integrated development environment (IDE), supporting collaborative and efficient code management. The backend services were implemented using Node.js, which provides a scalable, event-driven architecture suitable for handling real-time sensor data and API requests. For advanced data processing, particularly for anomaly detection and environmental computations such as dew point and VPD (Vapor Pressure Deficit), Python scripts were integrated into the backend pipeline.

    The mobile application was developed using React Native, enabling cross-platform deployment on both Android and iOS devices. This framework allowed rapid development and consistent UI/UX experiences across devices.
  - **Database Design** The system utilized MongoDB, a NoSQL database that supports document-based data structures. MongoDB was chosen for its flexibility and scalability, especially suitable for storing heterogeneous sensor data that varies in format and frequency. Each document represents a data packet from the sensors, including timestamp, temperature, humidity, and calculated parameters (e.g., dew point, VPD).
  - **System Architecture**The system architecture was

designed to support real-time data streaming, alert management, and user interaction. The sensor data is either simulated or gathered through MQTT-based communication and then processed by the backend. An alerting mechanism was implemented to notify users via the app interface and optional Line notifications when thresholds are breached (e.g., abnormal temperature, sensor malfunction, or missing data).

- **User Interface Design/User Flow**The mobile application interface was designed with a focus on simplicity, clarity, and accessibility, particularly for users in rural areas or those with limited technical knowledge. It includes a real-time dashboard displaying environmental data, providing users with immediate insights into the system's status. The interface also features color-coded alerts and device status indicators, making it easier for users to quickly identify any issues or abnormal conditions. Additionally, historical data is visualized through graphs, allowing users to track trends and changes over time. The application also includes settings for threshold configuration and device registration, enabling users to customize the system according to their needs and preferences.

  The design of user flow aims to define a clear sequence of steps for interactions between users and the system to ensure smooth and efficient application usage while minimizing confusion in each process. This is particularly important for users with varying levels of digital literacy, such as farmers in rural areas who may face limitations in accessing or understanding technology. Therefore, the system must be designed with a focus on simplicity, clarity, and ease of learning. To meet these goals, the system was developed and tested with user flows that cover key usage processes commonly encountered in everyday use ranging from initial steps such as user registration and password recovery, to adding locations and devices, viewing historical data, and managing personal account information. These processes can be categorized into five main flows, as follows

  UF1 – Sign-Up Flow, "As a new user, I want to sign up with a valid email and matching passwords so that I can create an account without errors."

  New users can register through a form that includes validation mechanisms to ensure the correctness of the entered email address and password. The system provides immediate feedback if any input errors are detected for example, if the passwords do not match thereby helping to prevent user mistakes and reduce confusion during the account creation process.

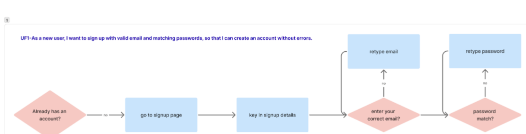

Fig. 4. UF1 – Sign-Up Flow

UF2 – Forgot Password Flow, "As a user, I want to recover access to my account if I forget my password, so that I can log in securely without creating a new account."

Users who forget their password can tap on the "Forgot Password" option and enter their registered email address. The system will then send a password reset link to the user's email, allowing them to create a new password and securely regain access to their account.

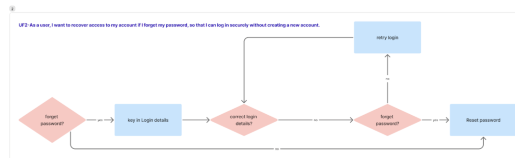

Fig. 5. UF2 – Forgot Password Flow

UF3 – Add Location Before Device Flow, "As a user, I want to be guided to add a location before adding a device, so that my devices are organized properly."

When using the system for the first time, if no location has been added, the user will be prompted to create a location such as a farm plot or greenhouse before they can register a sensor device. This process ensures that all devices are categorized by physical location, improving data organization and management efficiency.

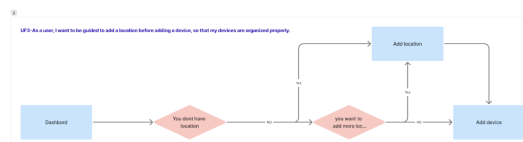

Fig. 6. UF3 – Add Location Before Device Flow

UF4 – View and Compare Data Flow, "As a user, I want to be able to compare data across multiple locations or view a single location, so that I can analyze trends more accurately before selecting a date range."

On the Statistics screen, users can choose to view data from specific sensors, either individually or by comparing multiple locations. The system will display the data in a graph format, with options to select the desired date/time range for further analysis based on the user's needs.

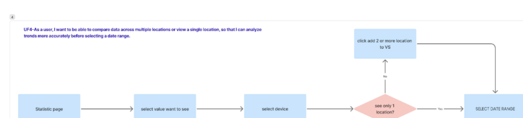

Fig. 7. UF4 – View and Compare Data Flow

UF5 – Edit Profile and Password Flow, "As a user, I want to update my profile and password, and be alerted if the new passwords don't match, so that I can securely manage my account information."

The user can access the settings page to update profile information such as name, email, and password. The system will validate and notify the user if the entered passwords do not match, in order to prevent errors and maintain account security.

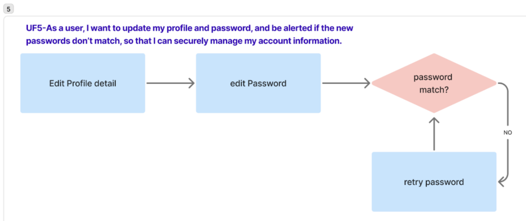

Fig. 8. UF5 – Edit Profile and Password Flow

- **Usability Testing**The objective of this usability testing is to assess the effectiveness, efficiency, and user satisfaction with the smart farming system developed based on Internet of Things (IoT) technology. The testing was conducted in a real-world context with actual users from rural areas in Chiang Rai Province, who represent a unique user group in terms of their technological literacy, familiarity with digital devices, and everyday usage patterns. The main focus of the testing was to evaluate whether the application could effectively meet the needs of the target users. This evaluation was based on their interaction with the User Interface (UI), their overall User Experience (UX), and the identification of any issues that might arise during real-world use such as the complexity of the setup process, clarity of sensor data displays, and the performance of the alert system. The results of the testing served as the basis for further refinement and development of the system. These refinements were guided by users' responses to various functions, their understanding of the application structure, ease of data access, and their ability to manage IoT device controls effectively. Ultimately, the goal was to ensure the application truly supports real-world use cases within the specific context of Thai farmers.
  - **1. Population** This study employed a qualitative research approach with the objective of developing and evaluating a smart farming system that utilizes Internet of Things (IoT) technology for environmental monitoring in real-world agricultural settings. The population targeted in this study consisted of actual users, specifically farmers or farm managers who have experience using IoT technologies. A total of six individuals from Chiang Rai Province were selected to participate.
  - **2. Sampling Method** The sample was selected using purposive sampling, based on the participants' roles and relevance to the smart farming ecosystem, particularly in the context of utilizing sensors for monitoring temperature and humidity in farming environments. The sample included two main groups: (1) users who operated smart farming systems through sensor devices alone, and (2) users who employed both sensor devices and software applications for controlling farming equipment.

    Key informants were identified through referrals from local personnel and experts familiar with the farming technology landscape. The selection criteria for participants included: (1) individuals who currently use or have previously used IoT-based smart farming sensors, and (2) individuals who are capable of providing in-depth information regarding their usage experiences and challenges encountered with the system.
  - **3.Scope of Testing** This study defined the testing scope to evaluate the usability performance of the smart farming application developed based on IoT technology. The testing was conducted through mobile devices running on both Android and iOS operating systems, focusing on assessing real-world user experience among farmers in rural areas. The evaluation covered all core components of the application related to controlling sensor-based systems. Emphasis was placed on assessing the system's responsiveness to actual usage scenarios, ease of access to functions, and comprehensibility for users with varying levels of technological proficiency. The scope of testing began with evaluating the user registration and login processes, including account creation and system access. It also covered the process of adding IoT devices, where users needed to select a device from a list, connect it to the system, and verify the device's status. The system was expected to display real-time sensor data such as temperature, humidity, and dew point accurately and allow users to configure device settings, including renaming devices or comparing data displays across different devices or locations. In addition, the testing assessed overall system status, including device battery levels, network stability, and the system's ability to issue alerts in case of malfunctions or abnormalities. The alert system evaluation focused on verifying the accuracy and stability of error notifications and the user's ability to manage notification volume. Historical data analysis was also a critical area, requiring the system to present historical trends by date, time, or device, and allow users to filter or summarize the data for efficient farm management insights.

    Lastly, the system settings were tested to verify whether users could easily adjust configurations related to personal information, security, language preferences, and notification settings. Any changes made were expected to take effect immediately without disrupting overall system functionality. This comprehensive testing scope aimed to provide an in-depth assessment of the application's usability, identify its strengths

– **4.Task Scenarios** The test scenarios in this study were designed to evaluate whether the IoT-based smart farming system effectively responds to user needs within real-world contexts. Each scenario was constructed to reflect actual tasks that users must perform when managing their farming systems and encompasses all critical components of the application, including functional operations, data presentation, and user interactions:

Scenario 1: Register and Log in,

The objective of this testing scenario was to evaluate the accuracy, stability, and efficiency of the registration and login process within the application. The goal was to verify that users could successfully create an account and access the main dashboard without encountering errors and within the expected time frame. The procedure began by instructing participants to open the "Welcome Back" screen, which serves as the main login interface. If the participant did not have an existing account, they were required to tap the "Sign Up" link located at the bottom of the screen to access the "Create Account" page. Each participant was provided with sample registration credentials, including the email Test01@gmail.com, password 123456, and confirmation password 123456. Once all fields were completed, participants tapped the "Sign Up" button to submit their information to the system.

The system was designed to automatically verify data accuracy, including email format validation and password confirmation. If all inputs were valid, the system created a new account and redirected the user to the main interface. During this process, several usability metrics were defined for data collection and performance evaluation. Effectiveness was measured using the Rate of Success, defined as the proportion of participants who could complete the registration process and reach the main screen without any critical errors. Efficiency was measured using the Time on Task, which recorded the time taken from tapping the "Sign Up" button until the main page was fully displayed. The performance benchmark was categorized into three levels: completion within $\leq$ 60 seconds was rated as excellent, 61–120 seconds as moderate, and $>$ 120 seconds as needing improvement.In addition, the Error Rate was observed to record issues such as invalid email input, mismatched passwords, or incorrect button selection. Participants who made no more than one error were considered to have met the usability criteria, while more than four errors indicated potential user interface (UI) design issues.

Furthermore, the System Response dimension was assessed by measuring the application's loading time after the user tapped the "Sign Up" button. A timer was used to capture the duration between data submission and the appearance of the main interface. The response performance was categorized as excellent if the system loaded within ≤ 3 seconds, acceptable between 3–5 seconds, and needs improvement if longer than 5 seconds. Any signs of lag, screen freeze, or unresponsive transitions between pages were also recorded to assess system stability.

Scenario 2: Add Device,

This scenario was designed to evaluate the usability and stability of the process for adding a new IoT device to the system. The primary objective was to verify whether users could successfully discover, select, and connect a new device to the application within the specified time and without critical errors. The testing procedure consisted of two main tasks: selecting an IoT device from the displayed list and establishing a stable connection between the selected device and the application.

At the beginning of the test, users were instructed to open the "Select Device" screen, where the system displayed two categorized lists: "MY DEVICES", which contained previously paired devices, and "OTHER DEVICES", which included new devices available within Bluetooth or Wi-Fi range. Users were required to tap the "Connect" button located next to the desired device name. Once selected, the system automatically navigated to the "Connect Device" interface, where the pairing process was visually represented using animated feedback. When the connection was successfully established, the system displayed the message "Connected" on the screen to confirm task completion.

Data collection in this scenario focused on three primary usability metrics: Effectiveness, Efficiency, and System Response, along with an additional Error Rate (ER) indicator to complement the quantitative analysis.

For Effectiveness, the Task Success Rate (TSR) was calculated to determine the proportion of participants who successfully completed the task without exceeding the predefined error threshold. The formula used was:

$$\text{TSR} = \left(\frac{\text{Number of users completed the task successfully}}{\text{Total number of participants}} \right) \times 100$$

A success rate above 80 percent was considered indicative of good usability. Errors were defined as user actions such as selecting the wrong device, double-tapping the "Connect" button, or failing to recognize the visual feedback indicating successful pairing.

For Efficiency, the Time on Task (ToT) was measured as the duration from the moment the user opened the "Select Device" screen until the system displayed the "Connected" status. The average completion time was calculated using the following formula:

$$\bar{T} = \frac{\sum T_i}{n}$$

where T_i represents the time in seconds taken by participant i, and n is the total number of participants ($n = 7$ in this study). The performance levels were classified as follows: $\leq$ 30 seconds = excellent, 31–60 seconds = acceptable, and $>$ 60 seconds = needs improvement.

The System Response Time (SRT) was calculated to measure the interval between the user's tap on the "Connect" button and the system's display of the "Connected" message. The calculation was performed using the following formula:

$$\text{SRT} = \frac{\sum R_i}{n}$$

where R_i represents the response time in seconds recorded for participant i, and n denotes the total number of users. The classification criteria were: ≤ 3 seconds = excellent, 3–5 seconds = moderate, and > 5 seconds = needs improvement. In addition, the Error Rate (ER) was computed to quantify the proportion of user errors relative to the total number of actions performed. The formula applied was:

$$\text{ER} = \left(\frac{\text{Total number of errors}}{\text{Total number of user actions}}\right) \times 100$$

All user interactions were recorded through screen-capture tools and observational logs. Quantitative data obtained from timers and error records were statistically analyzed, while qualitative observations—such as hesitation, confusion, or repeated actions—were examined using behavioral analysis techniques to identify potential user interface (UI) design issues. The integration of these quantitative and qualitative analyses provided valuable insights into both the system's technical performance and users' behavioral experiences during the device-adding process, forming a comprehensive basis for subsequent improvements in system usability and design efficiency.

Scenario 3: Device Monitoring,

This scenario was designed to evaluate the usability, accuracy, and responsiveness of the system's real-time monitoring feature, which allows users to view and interpret environmental data collected from IoT devices. The primary objective was to determine whether users could successfully access, read, and control the display of sensor data such as temperature, humidity, dew point, and vapor pressure deficit (VPD) within the specified time and without confusion or system delays. During the test, participants were instructed to log in to the application and access the "Device Monitor" menu from the main dashboard. The system then navigated to the "Device Status Overview" screen, displaying a list of connected IoT devices. Each user was asked to select one device to access the real-time monitoring screen, where the system presented sensor data in a card view layout. Each card represented one parameter (temperature, humidity, dew point, or VPD) and contained both the current value and a historical trend graph.

Data collection in this scenario focused on three main usability dimensions: effectiveness, efficiency, and system response, along with an additional measure of error rate to support quantitative analysis. Effectiveness was assessed by calculating the Task Success Rate (TSR), which represents the proportion of participants who successfully completed the monitoring task without exceeding the predefined error threshold. The TSR was calculated using the following formula:

$$\text{TSR} = \left(\frac{\text{Number of users completed the task successfully}}{\text{Total number of participants}}\right) \times 100$$

A TSR value higher than 80 percent indicated good usability. Errors in this context included failure to display parameter data, difficulty using toggle buttons, or inability to recognize the graphs on the screen.

Efficiency was measured through Time on Task (ToT), defined as the duration between the user's tap on the "Device Monitor" menu and the moment all four parameter cards were fully displayed. The average completion time was computed using:

$$\bar{T} = \frac{\sum T_i}{n}$$

where T_i represents the time taken by participant i, and n denotes the total number of users ($n = 7$). The performance scale was categorized into three levels: completion within 15 seconds was rated as excellent, between 16–30 seconds as acceptable, and above 30 seconds as needing improvement.

The System Response Time (SRT) was evaluated by measuring the interval between entering the monitoring screen and the full rendering of all data cards and graphs. The mean response time was calculated using:

$$\text{SRT} = \frac{\sum R_i}{n}$$

where R_i refers to the rendering time for participant i. The system performance was classified as excellent when the response occurred within 3 seconds, moderate for 3–5 seconds, and poor if the response exceeded 5 seconds.

To complement the usability evaluation, the Error Rate (ER) was calculated as the ratio of the total number of user errors to the total number of user actions, expressed as a percentage, using:

$$\text{ER} = \left(\frac{\text{Total number of errors}}{\text{Total number of user actions}}\right) \times 100$$

All interactions were recorded through on-screen capture tools and observation notes. Quantitative data such as task time, response time, and success rate were analyzed statistically, while qualitative behaviors such as hesitation, searching for buttons, or repeated toggling were examined using behavioral observation techniques.

Scenario 4: System Status,

This scenario focuses on enabling users to check the system status connected to IoT devices. It includes monitoring battery levels, connection stability, and device health. The system displays this information using color-coded boxes that clearly indicate different statuses, along with alerts when devices require attention such as sensor calibration or maintenance.

The operation steps in this scenario begin with the user accessing the main screen of the application. After successfully logging in, the user taps on the device display box shown on the main dashboard. The system will then navigate the user to the "Device Monitor" screen. From there, the user must tap on the device name, such as "Sensor IBS-TH3," to access the data display screen. Upon entering this screen, the system will display basic information such as WiFi Status, Battery Status, Data Status, and Device Health. Symbols are used to represent different statuses: green (normal status), yellow (low battery), and red (device issue), allowing users to quickly assess the situation. At the bottom of the screen, users will see a "Current Issues" section that displays alert messages about device problems, such as "Temperature sensor calibration required" or "Humidity sensor needs maintenance." If users wish to view past issues, they can tap the "Error History" tab at the bottom of the screen to access a list of previous error messages. Users can tap on each item to view more details.

Data collection in this scenario will be conducted through user behavior logging, time-on-task measurement for each step, and post-test interviews. Timing begins when the user taps the "System Status" menu and continues until they can identify the status of the device, such as a low battery or a sensor issue. If this process takes more than 10–15 seconds, it may indicate issues with data layout or comprehension of the symbols. The researcher will observe whether the user can correctly interpret the information from the color-coded boxes for example, whether they understand that red indicates an issue requiring immediate action. The researcher will also assess the user's ability to correctly tap the "Error History" button to access past error information. Any incorrect taps or failure to access the intended data will be recorded as an error

Scenario 5: Notification Testing,

This scenario focuses on enabling users to check the system status connected to IoT devices. It includes monitoring battery levels, connection stability, and device health. The system displays this information using color-coded boxes that clearly indicate different statuses, along with alerts when devices require attention such as sensor calibration or maintenance.

This scenario aimed to evaluate the usability, accuracy of user perception, and system responsiveness of the notification feature, which functions to alert users about device status, abnormalities, and environmental changes detected by sensors. The test focused on determining whether users could access, categorize, and manage notifications according to their severity levels (Critical, Warning, Info), as well as delete, filter, and handle notifications efficiently without errors and within the designated time frame.

The testing procedure began with users tapping the bell icon located at the top-right corner of the home screen to access the Notification Screen, where all alerts were displayed in chronological order. Each notification was visually categorized by both color and icon red for Critical (severe alerts), orange for Warning (moderate alerts), and blue for Info (general information). Users could tap the Filter button at the top of the screen to display only specific categories of notifications. The menu icon allowed users to select either Delete or Delete all, enabling them to remove individual or all notifications as needed.

The data analysis for the Notification Testing scenario was conducted using four primary quantitative indicators: Task Success Rate (TSR), Time on Task (ToT), System Response Time (SRT), and Error Rate (ER). These metrics were systematically applied to evaluate system usability and performance.

Effectiveness was assessed using the Task Success Rate (TSR), which measured the proportion of participants who successfully completed the notification-related task without exceeding the predefined error threshold. TSR was calculated using:

$$\text{TSR} = \left(\frac{\text{Number of users completed the task successfully}}{\text{Total number of participants}} \right) \times 100 \tag{1}$$

A TSR value greater than 90% indicated excellent usability.

Efficiency was measured using the Time on Task (ToT), defined as the time participants required to access and manage notifications. The mean completion time was calculated as:

$$\bar{T} = \frac{\sum_{i=1}^{n} T_i}{n} \tag{2}$$

where T_i represents the task duration (in seconds) for participant i, and n denotes the total number of participants. The interpretation criteria were:

- ≤ 7 seconds: excellent
- 8–10 seconds: moderate
- > 10 seconds: needs improvement

System Response Time (SRT) measured the duration required for the system to load and display all notifications after the user tapped the notification icon. The mean SRT was computed using:

$$\text{SRT} = \frac{\sum_{i=1}^{n} R_i}{n} \tag{3}$$

where R_i is the response time (in seconds) for participant i. Evaluation criteria were defined as:

- ≤ 3 seconds: excellent
- 3–5 seconds: moderate
- > 5 seconds: needs improvement

Error Rate (ER) was calculated to quantify the proportion of user errors relative to the total number of user actions performed during the task. ER was computed as:

$$\mathrm{ER} = \left(\frac{\text{Total number of errors}}{\text{Total number of user actions}} \right) \times 100 \quad (4)$$

These quantitative indicators provided a systematic approach to assessing the usability and reliability of the notification system. The calculated metrics were used to support further qualitative analysis based on user behavioral observations during testing.

Scenario 6: Statistics,

This scenario aimed to evaluate the usability, accuracy, and responsiveness of the system's statistical visualization module, which allows users to view and analyze historical data obtained from IoT sensors. The primary goal was to determine whether users could correctly access, filter, and interpret statistical graphs such as temperature, humidity, and dew point within the defined time frame and without operational errors. The test also examined how effectively users could use date range tools and distinguish multi-colored lines in the data visualization.

The testing procedure began with users accessing the Home Screen of the application and tapping the "Statistics" menu located in the middle tab of the navigation bar. Upon entering the "Statistics" screen, users could select a sensor (e.g., Sensor IBS-TH3) to view the associated greenhouse or plot (e.g., Greenhouse A, B, C). Once selected, the system displayed line graphs of key parameters, allowing users to compare environmental trends among different zones. Users could then define a time range by selecting the Start Date and End Date using the date picker tool, after which the system automatically generated the corresponding graph. Each parameter was displayed using a different color-coded line for example, a blue line representing temperature and a purple line representing humidity to help users visually differentiate multiple variables.

To ensure a comprehensive evaluation, four primary quantitative indicators were measured: Task Success Rate (TSR), Time on Task (ToT), System Response Time (SRT), and Error Rate (ER).

Effectiveness was assessed using the Task Success Rate (TSR), which measured the proportion of participants who successfully completed the assigned tasks (selecting sensors, setting date ranges, and viewing graphs) without exceeding the predefined error threshold. TSR was calculated using:

$$\mathrm{TSR} = \left(\frac{\text{Number of users completed the task successfully}}{\text{Total number of participants}} \right) \times 100 \quad (5)$$

A TSR value above 80% was considered indicative of good usability. During this scenario, several users encountered minor issues related to the responsiveness of the date picker and the interpretation of overlapping graph colors, which were documented as part of the usability analysis.

Efficiency was evaluated using the Time on Task (ToT), defined as the time required for participants to access and understand the statistical graphs. The average completion time was computed using:

$$\bar{T} = \frac{\sum_{i=1}^{n} T_i}{n} \quad (6)$$

where T_i represents the task completion time (in seconds) for participant i, and n denotes the total number of participants. The interpretation criteria were:

- ≤ 15 seconds: excellent
- 16–30 seconds: good
- > 30 seconds: needs improvement

In this scenario, the average ToT was approximately 24 seconds, which falls within the "good" performance category.

System Response Time (SRT) was measured to determine how long the system required to load and render the statistical graphs after the user selected a sensor or date range. The mean SRT was calculated using:

$$\mathrm{SRT} = \frac{\sum_{i=1}^{n} R_i}{n} \quad (7)$$

where R_i represents the system response time (in seconds) for participant i. The performance benchmarks were defined as:

- ≤ 3 seconds: excellent
- 3–5 seconds: moderate
- > 5 seconds: needs improvement

Error Rate (ER) was computed to quantify usability issues such as incorrect date selection, failure to load graphs, or misinterpretation of line colors. ER was calculated using:

$$\mathrm{ER} = \left(\frac{\text{Total number of errors}}{\text{Total number of user actions}} \right) \times 100 \quad (8)$$

All user interactions were recorded through screen capture and behavioral observation logs. Quantitative data from TSR, ToT, SRT, and ER were analyzed statistically, while qualitative observations such as hesitation during menu selection, confusion with the date picker, and visual interpretation challenges were analyzed thematically. This comprehensive analysis enabled the identification of areas for improvement, such as enhancing the visibility of the date picker and increasing the color contrast of graph lines to support users with limited technical experience.

Scenario 7: Export Data,

This scenario focuses on enabling users to correctly export data from IoT devices according to the intended purpose of use. The system must allow users to select the types of data to be exported, such as Temperature, Humidity, Dew Point, and Vapor Pressure Deficit (VPD), as well as specify the desired date range for retrieving historical data and choose a file format appropriate for their needs, such as CSV, PDF, or Excel. Users can access the data export screen through two main channels. 1)From the Device Monitor screen by tapping the "Export Data" button located at the top right corner of the data graph, 2) From the Statistics screen, where the export button is displayed below the trend graph for each time period. Once on the "Export Data" screen, users will find a form to select data types by tapping the checkboxes next to the desired categories, such as Temperature and Humidity. Next, they must select the date range for the data export by tapping the "Start Date" and "End Date" buttons to define the scope of the data.Then, users need to choose the file format for export by tapping on their preferred format: CSV (suitable for use in Excel), PDF (suitable for reports), or Excel directly.The system will then display a preliminary data preview in the "Export Preview" section, which includes the selected data types, start and end dates, file format, and approximate file size. After reviewing the information, users can tap the "Export Data" button to proceed with downloading or exporting the selected data file immediately.

In collecting data from this scenario, the researcher will observe user behavior from the moment they begin accessing the "Export Data" screen through the designated system pathways (from Device Monitor or Statistics), recording the total time taken to complete the process as well as the accuracy in selecting variables, dates, and file formats. If the user takes more than 30 seconds to complete all selections or makes incorrect taps or misses any steps, these will be recorded as errors for behavioral analysis. After the test, the researcher will ask users whether they understood the entire process particularly the selection of date ranges and data types and will also check whether users can correctly read and understand the contents in the "Export Preview" section. This reflects whether the system design is transparent and supports users in verifying accuracy before exporting.

– **5.System Usability Scale (SUS)** To quantitatively evaluate the usability and user satisfaction of the developed application, the study employed the System Usability Scale (SUS), a standardized and widely recognized tool developed by Brooke (1996). The SUS questionnaire consists of ten statements designed to measure the perceived usability of interactive systems. Each statement is rated on a five-point Likert scale, ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). The statements include both positively and negatively worded items to minimize response bias and cover different aspects of usability, such as ease of use, system complexity, user confidence, and consistency of functions.The ten items are as follows:

  ∗ **(1) I think I will use this app periodically,**
  ∗ **(2) I feel this app is so complicated that it needs to be simplified,**
  ∗ **(3) I think this app is easy to use,**
  ∗ **(4) I think I need the help of a technician to be able to use this app,**
  ∗ **(5) I find the various functions of this application are well integrated,**
  ∗ **(6) I think there are a lot of inconsistent things in this app,**
  ∗ **(7) I think most people can learn this app quickly,**
  ∗ **(8) I find this app very complicated to use,**
  ∗ **(9) I feel confident to use this application well, and**
  ∗ **(10) I need to learn a lot of things first before using this app**

  For scoring, each participant's response was converted into a numerical value following the standard SUS scoring procedure. For positively worded items (Items 1, 3, 5, 7, and 9), the adjusted score is obtained by subtracting 1 from the selected rating. For negatively worded items (Items 2, 4, 6, 8, and 10), the adjusted score is calculated by subtracting the selected rating from 5. Each item yields a score between 0 and 4, and the total raw score of all ten items ranges from 0 to 40. The raw total is then multiplied by 2.5 to produce the final SUS score, resulting in a possible range of 0–100 points.

  A higher SUS score indicates a higher perceived usability, with 68 points generally accepted as the benchmark for "acceptable usability." Scores above this threshold suggest that the system is easy to use and well-designed, while scores below 68 indicate that usability improvements are required. This scale thus provides a reliable and standardized means of quantitatively comparing user satisfaction and system usability across participants.

– **6.Utilizing Results** The researcher will analyze the data obtained from the questionnaires and in-depth interviews to guide system improvements, making it more suitable for the users' context particularly in terms of user interface (UI) design and user experience (UX). Feedback indicated a need for clearer presentation, reduced complexity, and support for users with varying levels of technological literacy. From the qualitative data, users reflected issues and suggestions such as unclear notifications, complex menu structures, and limitations in viewing historical data. As a result, the system has been improved to better meet user needs in both efficiency and satisfaction. Following these improvements, a re-testing phase is planned to evaluate whether the implemented changes effectively resolve the identified problems and to ensure that the system performs reliably under real-world conditions in agricultural settings. The re-testing will be conducted using the same task scenarios in order to compare usability performance and user satisfaction between the previous and improved versions. Additional data will also be collected to analyze trends for continuous system development.

Moreover, to ensure that the system can serve a wider user base and reduce inequality in access to technology, the researcher has also developed various forms of supporting instructional materials. These include a User Guide and an Instructional Video tailored to the target user group, using simple and clear language, real-life examples, and a design compatible with basic devices such as Android and iOS smartphones or tablets devices that are widely accessible to farmers. The development of these support materials not only helps reduce technological knowledge barriers but also empowers users to operate the system with greater confidence, minimizing the need for technical assistance in the long term. Through a structured feedback loop starting from collecting in-depth data through real-world usage, iteratively redesigning the system, and preparing user support materials the researcher has been able to create an innovation that is not only technically stable and efficient but also genuinely connected to the real-life context of Thai farmers. This approach aligns with the principles of Human-Centered Design and supports the sustainable adoption of digital technology in the agricultural sector.

- **Analysis**
  - **Quantitative Analysis** The quantitative data gathered through the survey such as ratings on system performance, ease of use, and user satisfaction will be analyzed using descriptive statistical methods. Measures of central tendency, particularly the mean, will be used to identify general trends in user satisfaction and system performance. Frequency distributions will help highlight recurring issues, such as problems with sensor connectivity or interface complexity. Additionally, cross-tabulation will be applied to explore relationships between demographic characteristics such as years of farming experience and the types of issues reported by users, providing insights into how different user groups interact with the system.
  - **Qualitative Analysis** The qualitative data, including user feedback on specific pain points, will be analyzed using thematic analysis. This process involves coding the responses to identify recurring themes related to issues such as sensor connectivity, user interface challenges, and other technical or usability concerns. The feedback will then be categorized based on the severity and frequency of reported problems, enabling alignment with targeted design recommendations. Actionable insights will be extracted from open-ended responses to gain a deeper understanding of the real-world experiences and needs of IoT system users in the farming context.
  - **Data Triangulation** The analysis will use data triangulation, where both the quantitative survey data and qualitative feedback will be compared and cross-verified to ensure that the findings are comprehensive and accurate. This will help in identifying consistent issues across different data types and user groups.
  - **Synthesis of Findings** Once the data has been processed and analyzed, the findings will be synthesized to identify the most critical issues impacting IoT system performance and user satisfaction. The final analysis will summarize the data in a way that supports the development of tailored recommendations for improving system stability and usability.
  - **Statistical Significance and Interpretation** To ensure that the identified patterns and findings are statistically significant, appropriate tests will be performed to verify the relationships between different variables, such as user demographics and reported system issues. The results will be interpreted with a focus on their practical implications for IoT system design, particularly in terms of addressing user pain points and improving overall user experience.

Conclusion This chapter presented a comprehensive methodology for the design, development, and evaluation of an IoT-based smart farming system grounded in Human-Centered Design (HCD) principles and supported by a Design-Driven Research framework. The research employed both quantitative and qualitative approaches to ensure that the final system truly responds to the needs, expectations, and constraints of its intended users farmers in rural Chiang Rai Province.

The design process followed five key stages: identifying user needs, defining system requirements, designing user-oriented solutions, developing a functional prototype, and conducting systematic testing through real-world usage scenarios. Each stage was informed by empirical data gathered through online surveys, in-depth interviews, and iterative usability testing. Core features such as sensor reliability, real-time alerts, historical data visualization, and user-friendly interfaces were prioritized to accommodate limited digital literacy and unstable infrastructure commonly found in rural farming areas.

System testing both functional and usability confirmed the practicality and relevance of the prototype, while the integration of the System Usability Scale (SUS) provided standardized metrics for user satisfaction. Analysis methods included descriptive statistics, thematic content analysis, and data triangulation to ensure a robust understanding of user experience and technical performance.

By integrating user feedback at every stage of the development process, this study demonstrates a participatory and evidence-based approach to innovation in agricultural technology. The methodological rigor applied here not only supports the creation of a contextually appropriate system but also lays the foundation for ongoing improvements, broader adoption, and sustainable use of IoT technology in Thai agriculture.

IV. IMPLEMENTATION

A. System Overview

The developed real-time environmental monitoring system is designed to support smart farming practices by collecting, analyzing, and visualizing environmental data primarily temperature and humidity. The system architecture is composed of three main layers: (1) simulated environmental data generated through a trained API using synthetic datasets, which serves as a substitute for physical IoT sensors; (2) a cloud-based backend responsible for real-time data processing, anomaly detection, and storage; and (3) user-facing applications available on both mobile and web platforms. The simulated data is derived from pre-trained models or datasets that emulate realistic environmental conditions, allowing the system to function independently of physical hardware during the development and testing phases. This layered architecture enables seamless data transmission, responsive analysis, and intuitive visualization, thereby supporting farmers in decision-making and environmental management for optimal crop growth.

B. Design Goals

The system is developed with five key design goals. First, reliability is prioritized through the implementation of robust hardware components and fail-safe mechanisms to ensure uninterrupted operation even in challenging field conditions. Second, the system is built with scalability in mind, capable of supporting up to 100 IoT devices simultaneously without performance degradation. Third, usability is emphasized by providing intuitive and accessible interfaces for both mobile and web applications, enabling farmers to interact with the system effortlessly. Fourth, the system promotes energy efficiency by utilizing solar-powered modules with battery backups, allowing for sustainable operation in remote agricultural areas with limited access to electricity. Lastly, data security is ensured through end-to-end encryption during data transmission and secure storage protocols using TLS/SSL, protecting sensitive agricultural data from unauthorized access or tampering.

C. System Architecture

The architecture diagram of the IoT-based smart agriculture system consists of three interconnected components that work together to support environmental monitoring and decision-making. These components include the sensor devices, the backend infrastructure, and the database system.

- **Sensor Device** Located on the left side of the diagram, the sensor device represents the initial stage of the system where data collection begins. In a real-world deployment,

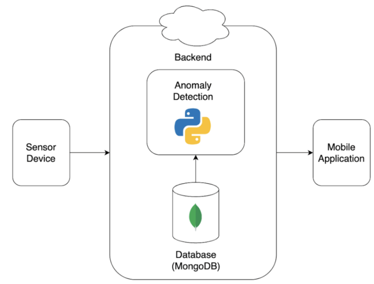

Fig. 9. ST1-System Architecture

these devices would be physically installed in agricultural fields to continuously monitor key environmental parameters. These include temperature, humidity, carbon dioxide concentration, electrical conductivity (EC), and pH levels. The sensor devices are configured to transmit the collected data to the backend system at defined time intervals using the MQTT (Message Queuing Telemetry Transport) protocol, which is lightweight and suitable for constrained environments. In the simulation environment of this project, actual sensors are not used; instead, synthetic sensor data is generated using Node.js. This approach allows the development and testing of system functionalities under conditions that approximate real-world sensor output, without requiring physical hardware during early stages.
- **Backend** The backend component, represented at the center of the system architecture diagram by a cloud-shaped and rectangular framework, serves as the core processing unit of the IoT-based smart agriculture system. This layer is responsible for receiving and handling data transmitted from the sensor devices, managing storage, executing data analysis functions, and supporting real-time services for the frontend applications. One of its primary roles is to ensure seamless data transmission and processing, facilitating responsive and data-driven decision-making for end users.

  Within the backend, the Database Subsystem is located at the lower part of the architectural frame. The system adopts MongoDB as its primary database due to its efficiency in handling large volumes of time-series data, which is particularly important for sensor applications that generate timestamped readings continuously. The database stores both current and historical sensor data to support analytics, system monitoring, and visualization. The diagram uses a green MongoDB icon to denote this component and its critical role in data management.

  Positioned above the database in the architectural stack is the Anomaly Detection module, a key component in enhancing system intelligence and reliability. Developed using Python, as indicated by the Python icon in the diagram, this module is integrated with MongoDB through a data retrieval pipeline that enables direct access to sensor data for analysis. The anomaly detection system utilizes a hybrid approach that combines rule-based detection based on predefined environmental thresholds with machine learning models trained to recognize patterns and deviations indicative of anomalies. This dual-method framework enables the system to detect conditions such as excessively high or low temperatures, irregular humidity levels, or faulty sensor behavior. When such anomalies are identified, the module generates real-time alerts that are transmitted to the mobile application layer, allowing timely notifications and enabling farmers to respond promptly to potential risks. This proactive monitoring capability contributes significantly to improving operational reliability and safeguarding agricultural productivity.

- **Mobile Application** Located on the right side of the system architecture diagram, the mobile application serves as the primary user-facing component of the IoT-based smart agriculture system. It is developed using React Native, allowing the application to run seamlessly across both Android and iOS platforms. This cross-platform capability ensures broader accessibility and ease of deployment for diverse farming communities.

  The application communicates with the backend through RESTful APIs for standard data retrieval and WebSockets for real-time updates. This dual-channel communication enables the application to present live sensor data and deliver instant notifications. The user interface includes an intuitive dashboard that displays essential environmental information such as temperature, humidity, and other sensor readings through interactive graphs and visual indicators. In the event that the anomaly detection module identifies irregularities such as extreme environmental conditions or sensor faults, notifications are immediately shown to the user. Furthermore, users are provided with options to configure alert thresholds, view historical trends, and manage sensor devices through the application interface, making it a comprehensive tool for remote monitoring and system control.

  The data flow within the system, as illustrated in Figure ST1 – System Architecture, is represented by directional arrows that trace the movement of data across system components. Simulated or real sensor data is first transmitted to the backend component (depicted by arrows pointing from the left side of the diagram toward the central backend module). Upon arrival, this data is stored in the MongoDB database, which is optimized for handling time-series sensor readings.

  The Anomaly Detection module developed in Python and positioned above the database layer then accesses this stored data for further analysis. This interaction is indicated by an upward arrow from the database to the anomaly detection block in the diagram. The module analyzes the data using both rule-based logic and machine learning models to detect abnormal environmental conditions or sensor faults.

  Once an anomaly is detected or routine analysis is completed, the results, including alert messages and sensor status updates, are transmitted to the Mobile Application component. This is illustrated by arrows extending from the backend (center) to the application interface on the right side of the diagram. This data flow ensures that end users receive up-to-date and actionable information in real time.

  Several technologies work together to support this architecture. The backend leverages Node.js and Express.js for managing APIs and simulating sensor data. The MongoDB database is used for its capacity to store time-series sensor readings efficiently. Python is employed for implementing the anomaly detection system due to its rich ecosystem of machine learning libraries such as Scikit-learn and TensorFlow. Finally, React Native enables the development of a performant mobile application with a consistent user experience across platforms.

D. Software Design

The software system is designed to simulate and manage sensor data for a smart agriculture system. Using JavaScript, the system generates synthetic values for a variety of sensor types, including temperature, humidity, CO2, EC, pH, dew point, and VPD. These values are transmitted to the backend at regular intervals, with the added functionality of introducing simulated errors such as sensor malfunctions. This simulation allows the system to mimic real-world conditions and test its response to different environmental scenarios.

Once the sensor data is generated, it is sent to the backend for processing and storage. The data is stored in a MongoDB database, where each sensor reading is associated with a unique device and sensor identifier, referred to as the device ID and sensor ID. The data follows the ISO 8601 standard for timestamps, ensuring accurate tracking and historical reference. The 'sensor-data' collection in MongoDB stores the raw sensor readings, while additional metadata, such as sensor type and operational status, is managed in the 'sensors' and 'devices' collections. This organization facilitates efficient data retrieval and analysis.

A critical component of the backend is the anomaly detection module, which monitors incoming data to identify any deviations from expected values. The system employs both rule-based detection, utilizing predefined thresholds, and AI-based methods for more sophisticated pattern recognition. When an anomaly is detected, the event is logged in the 'anomaly-logs' collection, where it is associated with key details such as the type of metric, actual value, expected range, detection method, and severity. These logs are crucial for both troubleshooting and improving system performance. Notifications are generated whenever an anomaly is detected, and these notifications are stored in the 'notifications' collection, along with references to the affected user and device.

The backend also provides a REST API, enabling access to both real-time and historical sensor data, anomaly logs, and notification statuses. This API facilitates seamless communication between the backend and frontend components. To support real-time updates, the system uses WebSockets, ensuring that data is instantly transmitted to the user interface without the need for constant refreshing.

On the frontend, the mobile application is responsible for displaying the sensor data in an interactive and user-friendly format. The app utilizes dynamic line charts to visualize the sensor readings, and users can filter the data by different parameters, such as zone, device, and sensor type. The mobile application also delivers real-time alerts to users when an anomaly is detected, providing them with immediate notifications. Additionally, the app allows users to configure notification preferences and adjust other settings, ensuring that it meets the specific needs of each user. The user interface is designed with responsiveness and clarity in mind, ensuring a seamless experience on mobile devices. The combination of a robust backend, real-time data handling, and an intuitive frontend provides an effective tool for farmers to monitor and manage their agricultural systems.

E. Database Design

The database design for this system utilizes MongoDB, a NoSQL database known for its flexibility and scalability, particularly in handling time-series data, which is essential for sensor-based applications. MongoDB's document-oriented structure allows for efficient storage and retrieval of sensor data, making it an ideal choice for this application.

The primary collection within the database is sensor-data, which stores the raw sensor readings from various devices in the smart agriculture system. Each document in the collection represents a unique sensor reading, and it contains several key fields that define the environmental parameters being monitored. The sensor readings include temperature, humidity, CO2, EC, pH, dew-point, and VPD values, which are essential for monitoring the health and conditions of the agricultural environment. These values are collected in real-time or simulated at regular intervals.

The sensor-data collection also includes two critical tags: Device-id and Location, which allow for the organization and categorization of the data. The Device-id is a unique identifier for each IoT sensor device, ensuring that the data can be traced back to a specific device within the system. The Location tag is used to associate sensor data with a particular geographical or environmental zone, enabling farmers to monitor different areas of their fields or greenhouses.

In addition to the sensor readings, each document includes a Timestamp field, which records the exact time the data was collected. The timestamp follows the ISO 8601 format, ensuring consistency and compatibility across time-based records. This standardized format facilitates accurate historical tracking and makes it easier to analyze trends over time. Overall, the database design is structured to support efficient data storage, retrieval, and analysis, making it well-suited for real-time monitoring, anomaly detection, and data-driven decision-making in the context of smart agriculture. The use of MongoDB enables scalability, allowing the system to handle large volumes of sensor data as more devices are added to the network.

F. Data Flow

The data flow within the system begins with the generation of simulated sensor data by virtual IoT devices. Each device is linked to a specific user via a user-id in the devices table, and these devices are virtual representations of actual physical IoT devices. Devices are characterized by attributes such as device-name, location, wifi-status, battery-level, and operational-status, which provide important context for the management and monitoring of devices in the system. Each device contains one or more sensors, which are defined in the sensors table. Each sensor is associated with a specific device through the device-id and includes metadata such as sensor-type (e.g., temperature, humidity, pH), sensor-status, and timestamps for creation and updates. This structure allows for efficient tracking of the status and performance of each device and sensor.

As the system operates, each sensor generates regular readings, such as temperature or humidity values. These readings are stored in the sensor-data table, where each entry contains the following key fields: sensor-id (linking the data to the specific sensor), timestamp (indicating when the data was recorded), and value (the actual measured value). The backend system continuously monitors these sensor readings to detect abnormal patterns or values, a critical aspect of the system's anomaly detection process. The anomaly detection system utilizes both rule-based and AI-driven algorithms to identify unusual patterns. When an anomaly, such as a temperature reading exceeding safe thresholds, is detected, a record is created in the anomaly-logs table. This table includes crucial information such as sensor-id (identifying the source sensor), timestamp (indicating when the anomaly occurred), metric-type (the type of data affected, e.g., temperature), actual-value (the anomalous reading), expected-range (the normal range for the metric), detection-method (the method used to detect the anomaly), severity (the severity level of the anomaly), and status (indicating whether the anomaly is resolved or unresolved).

To keep users informed, the system generates notifications stored in the notifications table. Each notification is associated with a user-id (the recipient), a device-id (indicating the source of the alert), a notification-type (e.g., anomaly alert or device offline), a message detailing the issue, and a status field to indicate whether the notification has been read or acknowledged.

The frontend interface (accessible via web or mobile) allows users to view real-time data, historical trends, anomaly history, alerts, and manage devices and sensors. Real-time data is pulled from the most recent entries in the sensor-data table, while historical trends are based on time stamped sensor readings. Anomaly history is retrieved from the anomaly-logs table, and alerts and messages are obtained from the notifications table. Additionally, the system maintains a weather table, which stores contextual environmental data such as temperature, humidity, and a weather-description. This table enhances the system's analytical capabilities, allowing for the correlation of environmental conditions with sensor data or anomalies. For example, poor air quality or high humidity levels can be linked to specific sensor readings or abnormal patterns, providing users with a more comprehensive understanding of the conditions affecting their agricultural environment.

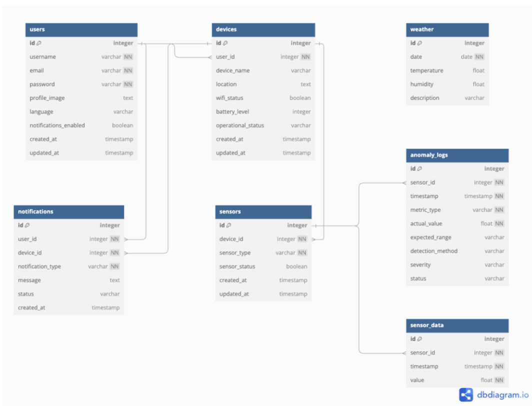

Fig. 10. ST2-ER-Diagram

G. Anomaly Detection Design

The anomaly detection subsystem acts as a sentinel layer to preserve data reliability and protect crop outcomes in an IoT-based smart agriculture environment. It adopts a two-tier strategy: a first tier of rule-based thresholds to capture clearly defined operational violations and a second tier of machine-learning models to identify subtler, multi-variable deviations that rules alone may miss. The design objective is to maximize detection sensitivity (recall) while controlling false alarms (precision) to fit real-farm workflows and minimize cognitive load for users with varying levels of digital literacy. Data governance principles—schema enforcement, timestamp integrity, and device provenance—are applied throughout to ensure that detection outputs remain interpretable, auditable, and actionable. The evaluation protocol and quantitative outcomes for this subsystem are detailed in Chapter 5, while this section focuses on the design and implementation choices that underpin those results.

- **Rule-Based Thresholds**
  The first layer implements domain-informed thresholds that provide immediate, explainable alerts for conditions with clear agronomic or operational implications. Representative examples include:
  - Temperature below 15 °C or above 35 °C,
  - Relative humidity below 40% or above 90%,
  - Vapor Pressure Deficit (VPD) below 0.5 kPa as an indicator of disease risk,
  - Dew point within 2 °C of ambient temperature as a proxy for condensation and mold formation.

  Equipment-centric rules monitor:
  - Supply voltage outside the range 2.8–3.5 V,
  - Sensors that remain constant for more than one hour,
  - Abrupt temperature shifts within a ten-minute window,
  - Data gaps longer than two hours.

  These rules are codified with explicit justifications and safety margins, versioned as policy, and executed before any model inference. This combination yields fast, transparent alerts and serves as a protective gate that either resolves an event outright or forwards it to the model tier when the signal is ambiguous or falls within a gray zone.
- **Anomaly Detection Models** The second layer employs machine-learning methods to capture non-linear and context-dependent patterns in time-series sensor data that cannot be robustly enumerated as rules. Several families were considered: Isolation Forest for partition-based separation of rare patterns, Local Outlier Factor (LOF) for density-based neighborhood comparisons, One-Class Support Vector Machine (OC-SVM) for boundary learning around nominal behavior, and ensemble strategies that combine complementary detectors. Selection criteria emphasized risk sensitivity, control of false positives, compute and memory footprint on the target infrastructure, ease of deployment and calibration, and predictable interaction with the upstream rule layer. Based on these criteria and comparative experiments, OC-SVM was adopted as the core detector, with ensembles reserved for scenarios where diversity brings measurable gains. All model choices, hyper-parameters, and training artifacts are tracked to ensure reproducibility; quantitative comparisons appear in Chapter 5.
- **Programming Languages and Libraries** The subsystem is implemented primarily in Python 3 to leverage its data-science ecosystem. Data preparation and feature engineering for time-series inputs use pandas and numpy, while scikit-learn provides reference implementations for the selected anomaly-detection algorithms and utilities for model selection. Visualization during development and analysis uses matplotlib (and, where appropriate, seaborn) to summarize distributions, residuals, and calibration curves. Models are serialized with joblib for deterministic deployment across environments. On the application side, a Node.js/Express backend orchestrates data ingestion, storage, and routing, exposing an internal API to a Python inference service that supports both micro-batch and streaming modes.
- **Model Evaluation Metrics** Performance is assessed using precision, recall, and F1-score to balance the competing costs of missed anomalies and false alarms in agricultural contexts. Precision quantifies the proportion of predicted anomalies that are truly anomalous, reflecting operator burden and alarm fatigue. Recall measures the proportion of true anomalies successfully detected, reflecting risk aversion to yield-threatening events. The F1-score, as the harmonic mean of precision and recall, serves as the principal summary for imbalanced datasets common

in anomaly detection. Where appropriate, threshold-free diagnostics such as ROC curves and the corresponding area under the curve (AUC) complement these metrics to understand separability across decision thresholds. Exact metric values and curves are reported in Chapter 5 under a fixed evaluation protocol.

- **Models Used and Their Comparative Behavior** To ensure that the chosen detector is justified beyond single-metric performance, each candidate model is evaluated along accuracy-adjacent measures, calibration behavior across operating points, stability under resampling, and failure modes on synthetic perturbations. OC-SVM, Isolation Forest, LOF, and ensemble variants are trained under identical data splits and preprocessing to isolate methodological effects from data leakage or configuration drift. Confusion matrices and error taxonomies are analyzed to attribute false positives to specific sensor modalities or environmental regimes, and to verify that recall gains are not driven by over-sensitivity to transient noise. The consolidated comparisons, including statistical confidence where applicable, are presented in Chapter 5.
- **Integration with the System** The inference capability is exposed as a stateless Python microservice reachable from the Node.js backend via REST for micro-batches and WebSocket for low-latency streams. New sensor records are validated and first evaluated by the rule layer; events that remain inconclusive or fall within a risk band are forwarded to the model layer for anomaly scoring. Inference results, confidence scores, and salient context (device identifier, location, and timestamps) are persisted in MongoDB and surfaced to downstream consumers, including an alerting center and a React Native mobile application. Idempotent request design, correlation identifiers, and structured logs enable reliable retries and end-to-end traceability across the pipeline.
- **Edge Case Analysis** Robustness to malformed or extreme inputs is addressed through an explicit data contract and pre-inference guards. The gateway enforces schema validation, unit consistency, and plausible value ranges; records with missing or infinite values, invalid types, or obviously unrealistic magnitudes are normalized, quarantined, or rejected according to policy. The system distinguishes transient gaps from persistent outages via sliding-window logic and applies back-pressure and retry strategies to avoid propagating partial state. For transparency, placeholder markers and standardized event codes record how each edge case was handled so that dashboards remain stable and operators can audit decisions. The breadth and results of edge-case testing are summarized in Chapter 5.
- **API Performance** Given on-farm connectivity constraints and the need for timely interventions, the API is engineered for low latency and high throughput under realistic load. The Python service employs lightweight request handlers, pooled model objects, and vectorized scoring to reduce per-call overhead, while the Node.js gateway batches compatible requests and applies connection reuse. Performance testing considers average and tail latencies, error rates, and sustainable throughput across representative batch sizes. Resource utilization is monitored to detect saturation and guide horizontal scaling policies. Detailed benchmarks and percentiles are reported in Chapter 5.
- **Performance Distribution** To avoid misleading conclusions from aggregate scores, the subsystem's behavior is examined across strata defined by device type, microclimate, time of day, and season. Stratified summaries reveal whether performance is concentrated in a narrow slice of conditions or is consistently achieved across the operational envelope. Where heterogeneity is observed, feature attribution and ablation analyses help identify inputs that drive instability, informing targeted rule refinements, sensor maintenance policies, or localized model recalibration. The corresponding distributional plots and narratives are included in Chapter 5.
- **ROC-AUC Analysis** Threshold-independent discrimination is assessed using ROC curves and AUC to compare detectors without committing to a specific operating point. This complements precision/recall analysis by characterizing separability when the decision threshold is tuned for different risk tolerances or alert budgets. The ROC framework also supports sensitivity studies in which thresholds are adjusted to match farm-specific constraints, such as staffing levels or crop susceptibility windows. Comparative ROC-AUC results for all candidates, along with guidance on threshold selection per deployment profile.
- **Notification and Escalation** Alerting translates detection outcomes into timely, actionable guidance for practitioners. Events are classified into at least two urgency levels: low-priority alerts for conditions that warrant monitoring or routine maintenance and high-priority alerts for safety-critical or equipment-fault scenarios requiring immediate action. Delivery channels include in-app banners and push notifications, with optional LINE/SMS routing configured per user preferences and connectivity realities. Each alert carries contextual evidence and recommended first-response actions, while all events are durably logged for retrospective analysis and periodic reporting. This design closes the loop from sensing to intervention, ensuring that model outputs lead to meaningful operational decisions and that field feedback can inform subsequent tuning of rules and models.

H. User Interface Design

The user interface (UI) is designed to offer an intuitive and efficient experience, allowing users to easily monitor and manage the system. It includes both a comprehensive dashboard for web access and a mobile application that provides real-time data and alerts. The UI aims to present information in a clear, accessible format while ensuring that critical conditions are promptly communicated to users.

- **Dashboard Features** The dashboard provides a comprehensive real-time data display, utilizing color-coded alerts to quickly inform users of any issues within the system. The color-coding ensures that users can identify critical conditions at a glance, improving response times. In addition to the real-time monitoring, the dashboard includes graphical trends that display historical data for key environmental factors such as temperature, humidity, VPD, and dew point. These trends can be viewed over different time intervals, including hourly, daily, and weekly, allowing users to track changes and identify patterns over time. The hardware status section shows the operational status of each device, including battery percentage, Wi-Fi connectivity, and online/offline status. This information helps users maintain system efficiency and address issues such as connectivity problems. Additionally, the notification center aggregates anomaly alerts, ensuring that users are immediately notified of any detected irregularities.
- **Mobile App Features** The mobile application is designed to offer seamless real-time data visualization, optimized for smaller screens to ensure usability on mobile devices. Critical alerts are pushed to users via notifications, enabling quick responses to any urgent issues. The app includes user authentication features, allowing users to sign in, sign up, and manage their account with options to change or reset passwords for improved security. Multilingual support is incorporated into the app, offering language selection for users from different regions, making it accessible to a wider audience. The app also allows users to manage different zones within the system, providing an option to add or rename zones as needed. Device management features enable users to add new devices or rename existing ones, providing flexibility in system configuration. In addition to sensor data, the mobile app displays local weather information, offering contextual insights that may impact sensor readings or system performance. Users can export data in multiple formats for further analysis or record-keeping. The app also offers device status monitoring, helping users ensure that each device is functioning optimally and that any issues are promptly addressed.

I. Deployment Plan

The deployment plan outlines the steps required to ensure a smooth and efficient launch of the system. It covers both the cloud infrastructure setup for the backend as well as the deployment of the mobile application for end-users. The plan is designed to provide a scalable and reliable solution for both the backend services and the mobile application.

- **Cloud Setup** The Node.js backend server will be hosted on a cloud platform to ensure scalability and availability. The choice of cloud provider will be based on factors such as cost-effectiveness, reliability, and ease of integration with other services. For database management, MongoDB Atlas will be used as the cloud-based solution to store and manage the simulated IoT sensor data. MongoDB Atlas is a fully managed cloud database that provides automatic scaling, security features, and high availability, making it a suitable choice for this system. The backend will be integrated with the mobile application to facilitate the retrieval of real-time sensor data and anomaly alerts. This integration will allow users to receive timely updates and notifications, ensuring they can take appropriate action based on the sensor readings.
- **Application Deployment** The mobile application will be built and tested using React Native, leveraging the Expo framework for a smooth development process and cross-platform compatibility. The application will undergo rigorous testing to ensure that it meets performance and usability standards across both Android and iOS devices. Once the testing phase is complete, the Android version of the mobile app will be packaged into an .aab file, which will then be published on the Google Play Store via the Google Play Console. This process will include creating an app listing, setting up appropriate metadata, and submitting the app for review. Similarly, the iOS version of the app will be packaged into an .ipa file and submitted to the Apple App Store via App Store Connect. The app will undergo Apple's review process, and upon approval, it will be available for download on iOS devices. This deployment strategy ensures that the app will be accessible to users on both major mobile platforms, providing them with a reliable and user-friendly experience.

J. Security Considerations

In the design of the system, security is a critical component, ensuring that all data, communication, and user interactions are protected from unauthorized access and potential cyber threats. The system implements multiple layers of security to safeguard both user and sensor data, ensuring privacy and integrity at all stages.

- **Data Transmission** To ensure the secure transmission of data between the mobile application and the backend API, all communication is encrypted using HTTPS (TLS/SSL). This ensures that sensitive information, such as sensor data, user credentials, and anomaly alerts, is securely transmitted over the internet and protected from interception. The use of HTTPS also mitigates the risk of man-in-the-middle attacks by providing end-to-end encryption. Additionally, the API endpoints are secured to prevent unauthorized access, ensuring that only authorized users and devices can interact with the system's backend services.
- **Authentication** The system employs JWT (JSON Web Tokens) to authenticate users and control access to protected resources within the application. Upon successful login, users receive a token that must be included in subsequent API requests to access restricted endpoints. This approach ensures that users are properly authenticated and authorized before accessing sensitive data or performing actions within the system. For added security,

user credentials are securely stored in the database and are hashed using industry-standard encryption algorithms, such as bcrypt. This means that even in the unlikely event of a database breach, user passwords remain protected.

- **Data Privacy** User data, as well as sensor records, are securely stored in MongoDB Atlas, which is a fully managed cloud database platform that offers built-in security features, including IP whitelisting, access controls, and encryption at rest. This ensures that sensitive information is safeguarded against unauthorized access both at the database level and during storage. Additionally, the system follows best practices for data privacy, ensuring that user information is kept confidential and protected. The system can be configured to comply with local data protection regulations, such as the Personal Data Protection Act (PDPA) or the General Data Protection Regulation (GDPR), depending on the geographical location and requirements of the user base. These security considerations ensure that the system meets the highest standards of data protection and user privacy.

K. Security Considerations

The system's performance is evaluated based on several critical metrics to ensure reliability, efficiency, and responsiveness. Data latency is one of the key factors, and the system is designed to provide real-time updates with a latency of less than 1 second. This ensures that users can access the most current sensor data and receive timely alerts without delay. Additionally, system uptime is a crucial metric, with the system expected to maintain an uptime of 99.5% or greater under normal operating conditions. This ensures minimal disruption to services and guarantees that the system is available for use almost all the time. In terms of anomaly detection accuracy, the system is designed to achieve a detection accuracy of 95% or higher, with minimal false positives, ensuring that alerts are reliable and relevant. Finally, the dashboard load time is optimized to ensure smooth user experiences, with the system loading within 5 seconds on standard internet speeds. These performance metrics collectively ensure that the system operates efficiently, providing users with a reliable and responsive experience for monitoring environmental conditions and sensor data in real time.

L. System Demonstration

To effectively showcase the capabilities of the IoT system for real-time temperature and humidity monitoring, a comprehensive demonstration was conducted, simulating a farmer's usage scenario through the mobile application. The demonstration highlighted key features of the system, illustrating how it works in a practical setting. The first step of the demonstration involves the user logging in to the system using their registered email and password. Once logged in, users proceed to add a device (sensor) by connecting a sensor device to the system. Upon successful connection, the device's status is immediately updated to "Online" on the application, signaling that it is ready for use. The system then allows real-time monitoring of temperature and humidity data, which is continuously updated every 5 seconds via WebSocket, ensuring that the user has access to the most up-to-date information at all times. Additionally, the system enables users to view historical data, where they can select a specific time range (day, month, or year) to analyze trends in temperature and humidity. To demonstrate the system's anomaly detection capabilities, the temperature is manually increased above 45°C, which triggers an automatic alert. As soon as the anomaly is detected, a push notification is instantly sent to the user's mobile device, alerting them to the critical condition. For users who need to analyze data in more detail, the export functionality is available. By navigating to the "Statistic" page, users can press the "Export" button to download the historical data in CSV format for further analysis. Finally, the demonstration includes the ability to check device status, where users can view device-specific details such as battery level, Wi-Fi status, and notification history, ensuring that they can effectively monitor and manage their devices at all times. This demonstration effectively highlights the system's real-time monitoring, anomaly detection, data export, and device management features, providing users with a comprehensive understanding of how the system operates in a real-world scenario.

M. Reproducibility

To ensure that the IoT system can be effectively replicated, deployed, and utilized in different environments or contexts, a detailed set of guidelines and resources have been provided. These steps serve as a comprehensive roadmap for installing, configuring, and running the system, ensuring consistency and reliability in its deployment.

For the installation of the system, several software components are required. These include Node.js, which is essential for running the backend server and handling API requests, and MongoDB, which serves as the cloud database solution for storing sensor data. Python 3.10+ is needed for various backend processing tasks and anomaly detection, while React Native Expo is required to build and deploy the mobile application. React Native can be initialized using the command npx create-expo-app@latest.

Once the necessary software is set up, the next steps involve replicating the project setup. First, the project can be cloned from the GitHub repository by running the command git clone https://github.com/Sorasak-dev/iotapp.git followed by cd iotapp. After cloning the project, the dependencies for both the frontend and backend are installed using the npm install command. For the backend, a Python virtual environment is created by navigating to the backend directory and running python -m venv venv, followed by pip install -r requirements.txt. After the dependencies are installed, the backend server is started using the command node server.js. To launch the frontend, Expo can be initialized using npx expo start.

Testing the system involves sending abnormal sensor values, such as increasing the temperature beyond safe thresholds, to ensure the anomaly detection and alerting system functions properly. Results should be checked by verifying that real-time and historical data are displayed correctly on the dashboard, notifications are successfully received on the mobile device, and data can be exported in both CSV and PDF formats. When the system is properly deployed, it should operate stably 24/7, with a latency of less than 5 seconds for real-time updates. Anomaly detection accuracy should be at least 95%, minimizing false positives and ensuring that the system can accurately identify critical issues. Furthermore, historical data should be easily exportable in both CSV and PDF formats, and the system should provide accurate error notifications for any issues that arise, enabling users to resolve them promptly.

By following these installation and configuration steps, the system can be reproduced and deployed in various environments with the same level of functionality and reliability. The provided guidelines ensure that the setup process is straightforward, and that the system operates as expected in different contexts, delivering consistent performance and support for users.

V. RESULTS

A. Introduction

This chapter presents the results and discussion of the evaluation of the proposed IoT-based monitoring system, focusing on its usability, anomaly detection performance, and security. The objective is to assess the system's readiness for real-world deployment and to identify areas for improvement. Usability testing was conducted through scenario-based tasks with target users, covering functions such as registration and login, device addition, monitoring, system status, notifications, statistics, and data export. These evaluations provided insight into the effectiveness, efficiency, and user satisfaction of the interface, particularly for users with limited digital literacy in rural agricultural contexts.

In parallel, the performance of the anomaly detection module was assessed using multiple machine-learning models, including One-Class SVM, Isolation Forest, Local Outlier Factor, and ensemble approaches. Metrics such as accuracy, precision, recall, F1-score, ROC-AUC, confusion matrices, and edge-case robustness were analyzed to determine detection reliability, operational readiness, and areas requiring additional safeguards. API performance and scalability were also evaluated to ensure the system could handle real-time streaming data with low latency and high throughput.

Finally, the system's security and privacy measures—including end-to-end encryption, JWT-based authentication, and database access control—were assessed to ensure the protection of sensitive user data. Overall, this chapter integrates quantitative and qualitative findings to provide a comprehensive view of the system's strengths, limitations, and readiness for deployment, establishing a foundation for subsequent improvements and practical applications.

B. Usability with Target Users

Scenario 1: Register and Log in,

The results of Scenario 1, "Register and Log in," demonstrated that the system achieved high stability and user-friendliness. All seven participants were able to correctly enter their registration details and successfully log into the application without encountering any system errors. This represented a 100% task success rate, indicating a high level of effectiveness in the registration form's design, which featured clear input field sequences and well-positioned confirmation buttons that facilitated ease of use.

In terms of efficiency, 85.7% of participants completed the task without confusion regarding buttons or menus, and 28.6% were able to finish within 60 seconds. The overall average completion time was approximately one minute, which is considered good performance. Participants who took more than two minutes were primarily those unfamiliar with mobile applications; however, they were still able to complete the task independently without assistance. This reflects the design's strong support for self-learning interfaces, enabling new users to understand and complete the process intuitively.

Regarding system response, all users confirmed that the Sign Up screen loaded smoothly within three seconds, with no freezing or restarting issues during screen transitions. This responsiveness helped users feel confident in the system's reliability. Behavioral observations revealed that new users could understand the step-by-step process from opening the app to accessing the main interface without needing to consult a manual. This outcome indicates a high level of discoverability and aligns with the design principle of digital inclusion, particularly relevant to Thai farmers with limited digital experience.

In summary, the Register and Login task achieved a 100% success rate, with an average completion time of approximately 80 seconds and no operational errors observed. The results confirm that the system is stable, responsive, and ready for real-world use, serving as a model for designing intuitive user interfaces for individuals with limited digital literacy.

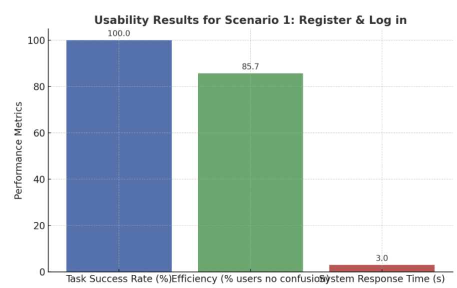

Fig. 11. Detection Scenario 1: Register and Log in

Scenario 2: Add Device,

The results of Scenario 2, "Add Device," revealed that most participants were able to complete the process successfully within a short time. The overall task success rate was 85.7%, which indicates good usability. All participants successfully added a device, and the system responded by displaying the status "Connected" within 3 seconds, demonstrating the stability of the IoT-device connection process. In terms of effectiveness, while most users understood the steps for pairing a device, a few participants particularly those unfamiliar with IoT systems experienced confusion when confirming the connection or waiting for the pairing to complete. Some reported uncertainty, saying, "I wasn't sure which button to press," or "The color changed too fast to notice." These findings highlight perceptual limitations in recognizing the connection status, which could be improved through clearer visual or auditory cues.

Regarding efficiency, the system exhibited fast response times ($\leq$ 3 seconds); however, the visual display of the "Connected" status was not sufficiently distinct. The low contrast between the background and the status text caused some users to spend extra time confirming the connection. It is recommended that the interface be refined by increasing the contrast ratio and adding a confirmation message such as "Device Connected Successfully", along with multimodal feedback (e.g., sound or vibration) to enhance user awareness.

In terms of satisfaction, users reported positive experiences, stating that once the device was connected, they felt confident and found the process easier to repeat. Experienced IoT users commented that the connection process was "faster than other apps they had used," while new users appreciated the system's simplicity but suggested adding brief explanations to improve understanding.

In summary, the "Add Device" task achieved an average success rate of 85.7%, with an average completion time of approximately 55 seconds and no major connection errors. The system demonstrated strong stability and accurate device detection. However, improvements are recommended to enhance the visibility of the "Connected" status and include textual guidance to assist novice users. These refinements are expected to improve both efficiency and satisfaction in subsequent testing rounds.

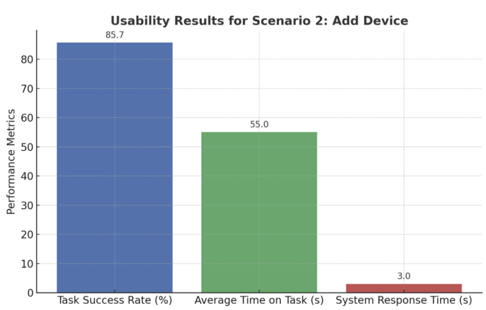

Fig. 12. Detection Scenario 2: Add Device

Scenario 3: Device Monitoring,

The results of Scenario 3, "Device Monitoring," showed that most participants were able to access complete data and clearly understand the presentation format of key parameters, including Temperature, Humidity, Dew Point, and Vapor Pressure Deficit (VPD). The overall task success rate was 85.7%, indicating a good level of usability.

In terms of effectiveness, all participants successfully accessed the device data screen (100%) without system errors. However, two users were unable to read all parameters because "the graphs did not appear" or "they did not realize they needed to toggle the graph on." This finding indicates a limitation in discoverability and icon clarity, suggesting that the visual cues for toggling graphs could be improved for better recognition. Regarding efficiency, the system performed with high responsiveness, loading all data within 3 seconds on average, and most users were able to view all key data within 15 seconds. Nonetheless, behavioral observation revealed that some users took longer because they were searching for graph control buttons or interpreting unfamiliar technical terms such as "VPD" or "Dew Point." This suggests the need for tooltips or explanatory labels to support users unfamiliar with IoT terminology.

In terms of satisfaction, users expressed that the background color and graphs were clear and visually appealing. However, some suggested increasing the contrast between graph lines for similar parameters such as humidity and temperature to make comparisons easier. Users also appreciated that the system provided real-time and uninterrupted updates, enhancing confidence in monitoring IoT device status.

In summary, the Device Monitoring task achieved an average success rate of 85.7%, with an average access time of approximately 13 seconds, indicating high efficiency and fast system responsiveness. Areas for improvement include enhancing the symbolic communication of toggle controls and adding unit explanations for technical metrics. These adjustments would improve data comprehension, especially among farmers or users with limited IoT experience.

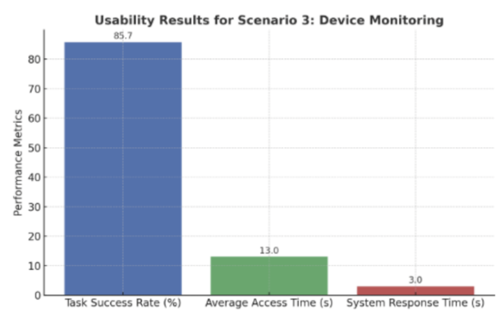

Fig. 13. Detection Scenario 3: Device Monitoring

Scenario 4: System Status,

The results of Scenario 4, "System Status," demonstrated that the system achieved the highest levels of effectiveness and efficiency. All participants successfully accessed the IoT system status without encountering any errors, including verifying the color-coded system indicators (green = normal, yellow = warning, red = error) and viewing the Error History, both of which operated flawlessly at a 100% success rate.

In terms of effectiveness, every participant immediately understood the meaning of the color-coded indicators without requiring further explanation. Participants commented that "green, yellow, and red are universally clear colors that anyone can understand," reflecting the success of universal visual coding principles. This visual design enhanced users' ability to interpret system conditions quickly and accurately.

Regarding efficiency, users took an average of only 9 seconds to open and review the system status and error history. No latency or loading issues were observed. Some participants were able to repeatedly open and close the error history without waiting for reloading, indicating low latency and high throughput an essential quality for real-time IoT monitoring systems. In terms of satisfaction, qualitative feedback indicated high user confidence and comfort with the system. Users mentioned that "the red alert color makes me feel confident that I'll notice problems immediately" and "I like the error history because it helps me understand what happened." This reflected a strong sense of control and trust in the system's operation. Several participants suggested adding brief error summaries or visual/audio alert enhancements (e.g., blinking lights or sound notifications) to increase responsiveness during field use.

In summary, the System Status task achieved a 100% success rate, an average completion time of 9 seconds, and zero errors. The system effectively displayed operational status with high accuracy, speed, and clarity, demonstrating strong operational readiness and suitability for real-time monitoring in rural agricultural contexts.

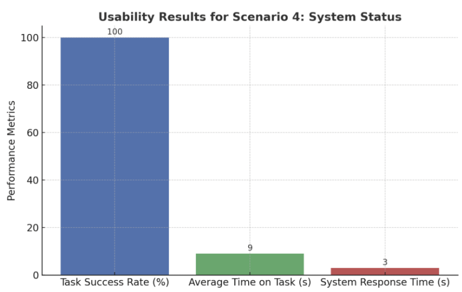

Fig. 14. Detection Scenario 4: System Status

Scenario 5: Notification Testing,

The results of Scenario 5, "Notification Testing," revealed that most users were able to quickly access the notification page and correctly understand the meaning of different alert levels Critical, Warning, and Info without confusion. The overall task success rate ranged between 85.7% and 95.2%, indicating excellent usability.

In terms of effectiveness, all participants successfully opened the notification interface and accurately recognized the alert types based on their color codes (red = critical, orange = warning, blue = information). Only one participant took longer than average due to uncertainty about the position of the Filter button. This indicates that the icon placement and touch target area of the button could be improved to enhance discoverability and ease of access.

Regarding efficiency, the system consistently loaded all notifications within 3 seconds, which meets the standard for mobile responsive systems. Most users were able to filter or delete notifications in under 10 seconds, with an average time of 7 seconds, indicating a smooth and efficient interaction flow.

In terms of satisfaction, post-test feedback indicated that users appreciated the color-coded and icon-based alert system, which helped them easily distinguish the severity of notifications. Many users commented that real-time alerts allowed them to quickly respond to environmental changes. Some suggested adding sound or vibration alerts for critical notifications, which would improve multisensory feedback and enhance user responsiveness, especially when users are not actively looking at their screens.

In summary, the Notification Testing task achieved an overall success rate of 85.7–95.2%, with an average completion time of 7 seconds and no critical errors. The system demonstrated excellent responsiveness, clarity, and high user satisfaction, particularly due to its intuitive color-coded design and direct message communication. These results confirm that the notification system is highly suitable for agricultural users who need real-time, easy-to-understand alerts.

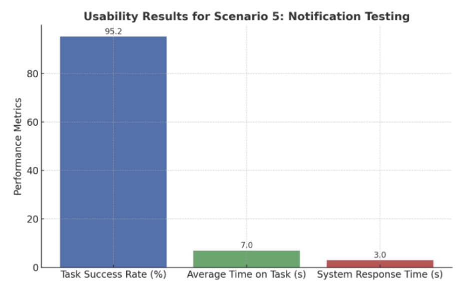

Fig. 15. Detection Scenario 5: Notification Testing

Scenario 6: Statistics,

The results of Scenario 6, "Statistics," show that most users were able to access the statistics page and correctly select the desired sensor/zone with a 100% success rate, indicating clear menu design and good user understanding. However, when users proceeded to the steps involving date-range selection and graph interpretation, several participants experienced difficulties particularly those unfamiliar with date-picking interactions or interpreting multi-colored line graphs. As a result, the overall task success rate decreased to 85.7%.

In terms of Effectiveness, the system successfully displayed all statistical graphs without system errors. However, two participants were unable to select the date range or reported that the date picker did not respond to their input. This issue was caused by the date picker button being insufficiently prominent and requiring multiple taps to activate. Some users also commented that "the graph colors are too similar" or that "the trends are not clear," indicating limitations in visual differentiation and clarity of graph design.

Regarding Efficiency, the average time spent viewing and interpreting the statistical data was 24 seconds, which is considered acceptable. Most users were able to understand relationships in the data such as the correlation between temperature and humidity but required additional time to read and compare the graph lines. Enhancing the color contrast and adding clearer unit labels or captions on the graphs would likely help reduce time and improve comprehension.

In terms of Satisfaction, most participants stated that the graphs were visually appealing but noted that "the colors are too similar." Several users suggested adding shortcut buttons, such as "Today," "Last 7 Days," or "Last 30 Days," to simplify selecting time ranges. Some participants who struggled with the date selection also reported that the system did not provide feedback when they tapped incorrectly, leaving them unsure whether the action was successful. This reflects a lack of feedback, suggesting that the system should include confirmation messages or small animations to indicate input response.

In summary, the "Statistics" task achieved an average success rate of 85.7% and an average completion time of 24 seconds, indicating good usability performance. The system responded quickly and was able to load graphs within three seconds. However, improvements are needed in the date selection mechanism (date picker) and the visual differentiation of graph colors to enhance comprehension and reduce interpretation time, especially for users with limited technological experience.

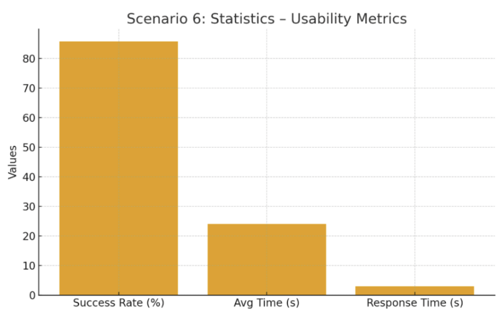

Fig. 16. Detection Scenario 6: Statistics

Scenario 7: Export Data,

The results of Scenario 7, "Export Data," demonstrated that this task achieved the highest level of usability performance among all scenarios. All participants (100%) were able to navigate to the statistics page, correctly select the appropriate sensor/zone, and choose the desired time range, indicating that the menu structure and component layout in the export interface were intuitive and easy to understand.

In terms of Effectiveness, all users successfully selected the correct file format (e.g., .CSV or Excel) and demonstrated a strong understanding of the Export Preview feature, with 85.7% showing complete comprehension. Only one participant expressed uncertainty, stating they were "not sure whether the graph shown in the Preview matched the actual exported file." This suggests the need to include a clarifying message such as "The preview displays only a portion of the data that will appear in the exported file" to enhance user confidence.

Regarding Efficiency, users completed the export process in an average of 20 seconds, which is considered excellent performance. Several participants noted that the steps for selecting sensors and time ranges were consistent and seamless, and that the Export Preview supported accuracy verification before saving the file, helping to reduce errors related to incorrect date-range selection.

For System Response, all participants were able to load the Export Preview within 3 seconds, with no loading delays or system freezes. The system responded smoothly to user actions, and no network-related issues occurred during data retrieval.

In terms of Satisfaction, most users described the process as "easy to understand, convenient, and fast," highlighting the clear export icon and easily accessible confirmation button. Additionally, having multiple file format options increased their sense of control and reflected the system's high level of flexibility.

In summary, the Export Data task achieved a 96.4% success rate, an average completion time of 20 seconds, and a 100% system pass rate. These findings confirm that the system is fast, intuitive, and effective for both users with technical backgrounds and novice users. The strong performance of this task reflects that the UI design of the IoT system adheres to user-centered design principles and is well-prepared for real-world use among Thai farmers.

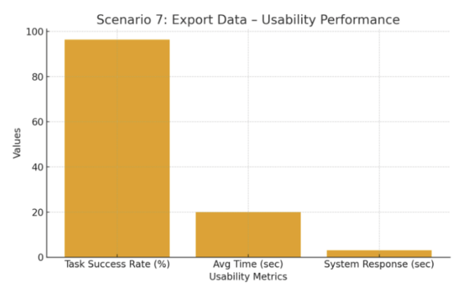

Fig. 17. Detection Scenario 6: Statistics

System Usability Scale (SUS) Evaluation Results,

The usability of the system was assessed using the 10-item System Usability Scale (SUS), with a total of five participants completing the questionnaire after interacting with the prototype IoT application. Based on the scoring method proposed by Brooke (1996), the overall average SUS score was 74.64 (SD = 13.18), with scores ranging from 57.5 to 97.5. When compared to the standard SUS benchmark where an average score of 68 is considered "acceptable usability" these results indicate that the system falls within the category of "Good Usability", approaching "Excellent", according to usability grading by Bangor et al. (2008). This implies that most users were able to interact with the system effectively, with minimal need for external assistance.

A more detailed analysis following the approach of Lewis and Sauro (2009) revealed that the system scored an average of 75.00 (SD = 14.43) on Usability (overall ease of use) and 73.21 (SD = 19.67) on Learnability (ease of learning). These closely aligned values suggest that users were able to learn how to use the system quickly and experienced a positive overall interaction. Qualitative feedback also highlighted that users found the menu structure easy to understand and that core functions such as data viewing, notifications, and data export operated smoothly, especially when brief instructions were provided beforehand.

However, some participants recommended improving the clarity of graphical displays and adding explanatory text to support interpretation of technical parameters. Examples include visual indicators for measurement units (°C, %RH) and explanations for index values such as VPD and Dew Point.

In summary, the SUS results demonstrate that the prototype system achieves "Good" usability across effectiveness, efficiency, and user satisfaction. The findings indicate strong potential for further development and adaptation of the system for general users in rural Thai agricultural communities.

C. Model Performance

The performance evaluation of the anomaly detection module aimed to verify the model's ability to distinguish between normal and abnormal states of environmental data streamed in real-time. The evaluation utilized a synthetic dataset reflecting seasonal patterns of temperature, humidity, and Vapor Pressure Deficit (VPD), including both threshold-based anomalies and subtle, hidden abnormal events. This approach allowed assessment of the model's sensitivity and specificity. Comparative testing was conducted using key algorithms, including Isolation Forest, Local Outlier Factor (LOF), One-Class SVM, and a weighted ensemble method. Standard performance metrics, such as Accuracy, Precision, Recall, F1-score, and ROC-AUC, were calculated on a previously unseen test set.

- **Overall Qualification Outcome** The anomaly detection subsystem was qualified on **September 2, 2025 at 09:58:38** under the fixed evaluation protocol described in Chapter 4. It achieved a system status of `CONDITIONAL_PASS` and an operational readiness level of `READY_WITH_MONITORING`.
  In practice, this indicates that model-level detection quality and API reliability meet baseline production criteria, while robustness to adversarial or malformed inputs still requires close observation and incremental hardening.
- **Comparative Model Performance.** Among the evaluated detectors—**Isolation Forest**, **Local Outlier Factor**, **One-Class SVM**, **Random Forest**, and an ensemble aggregator—the **One-Class SVM (OC-SVM)** delivered the most balanced performance on the held-out test set, with

$$\text{Accuracy} = 0.922, \quad \text{Precision} = 0.768,$$

$$\text{Recall} = 0.986, \quad \text{F1-score} = 0.863.$$

  Isolation Forest reached

$$\text{Accuracy} = 0.854, \quad \text{F1-score} = 0.713,$$

  indicating reduced sensitivity to subtle deviations.
  The **Ensemble** configuration achieved high recall (0.986) but materially lower precision, producing excessive false positives that would increase operator burden in field conditions.
  These outcomes justify the selection of **OC-SVM** as the production default, while reserving ensembles for deployments that can tolerate higher alert volumes.

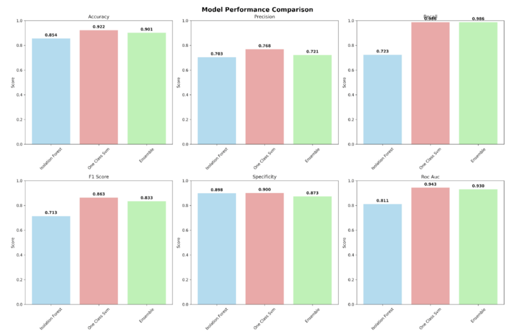

Fig. 18. Overall model comparison across metrics

- **Precision–Recall Behavior** Precision–recall curves confirm that OC-SVM maintains a favorable trade-off across operating points relevant to farm operations. At thresholds targeted for proactive intervention, recall remains near maximal with only a moderate precision concession, whereas Isolation Forest degrades more quickly as thresholds shift to increase sensitivity. This behaviour supports OC-SVM's use in scenarios where the cost of missing true anomalies (e.g., disease onset under low VPD) outweighs the cost of occasional false alerts.

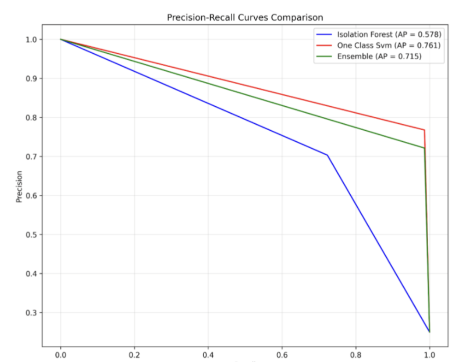

Fig. 19. Precision–Recall curves at relevant operating points.

- **Confusion Matrix Analysis** Confusion matrices reveal two dominant error modes. First, false positives cluster around transitional microclimate periods (rapid temperature changes within ten minutes), where signal volatility can mimic abnormal patterns. Second, a smaller band of false negatives appears during extended data gaps that were partially imputed, suggesting that imputation artifacts mask weak anomalies. Post-hoc inspection shows that incorporating rule-layer context (e.g., "data loss > 2

hours”) alongside model scores reduces these errors when alerts are fused at decision time.

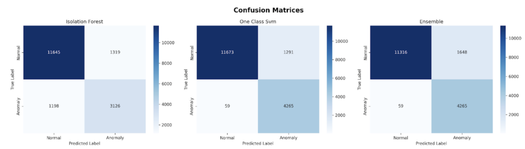

Fig. 20. Confusion matrices for evaluated models

- **ROC–AUC Discrimination.** Threshold-independent analysis further differentiates detectors. OC-SVM achieved ROC-AUC = 0.943, reflecting strong separability between nominal and anomalous instances. The Ensemble model attained ROC-AUC = 0.930, and Isolation Forest reached ROC-AUC = 0.811. These values corroborate the precision–recall findings and indicate that OC-SVM offers the most robust headroom for threshold tuning across farm-specific risk tolerances.

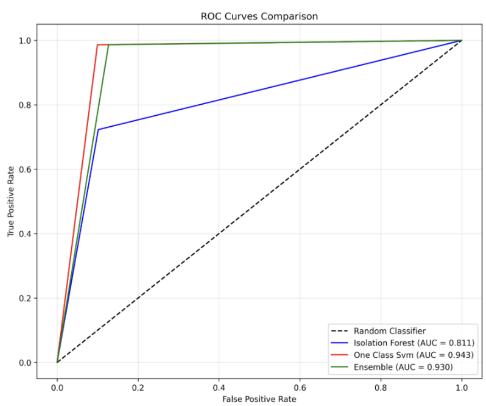

Fig. 21. ROC curves and AUC comparison.

- **Edge-Case Robustness.** In a targeted edge-case suite of 14 scenarios, the system successfully processed 6 cases (42.9%, below the ¿80% target). Failures were triggered by records composed entirely of None values, non-finite values, invalid data types, empty strings for numeric fields, and unrealistically large magnitudes beyond agronomic plausibility. While average inference latency during this suite remained low (0.0031 s) and memory usage negligible (0.16 MB), the error taxonomy highlights the need to strengthen pre-inference validation, unit checks, and quarantine workflows for suspect records.

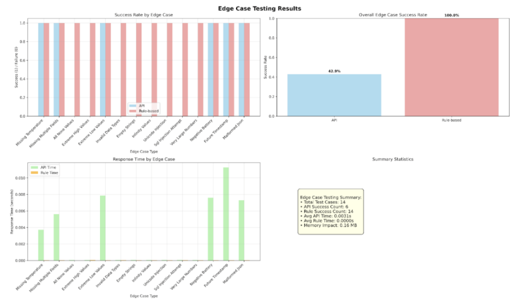

Fig. 22. Edge-case test outcomes and taxonomy

- **API Performance and Scalability** Service-level testing demonstrated average response time = 0.0034 s, P95 = 0.0038 s, and error rate = 0% under mixed micro-batch and streaming loads. Throughput scaling tests showed sustainable processing up to 15,898.96 samples/second at batch size 100, with stable resource profiles and no back-pressure events observed at target load. These results indicate sufficient performance margin for real-time use on typical farm deployments and room for horizontal scaling if sensor density increases

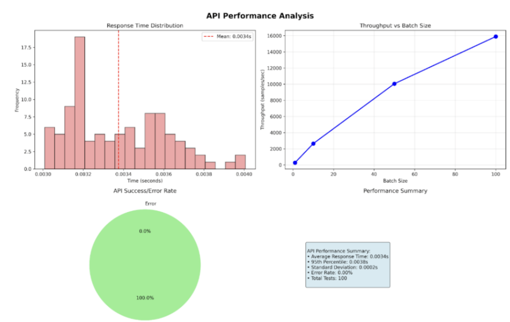

Fig. 23. API latency, P95, error rate, and throughput by batch size

- **Performance Distribution Across Conditions.** Stratified analyses by device, location, and time-of-day confirm that detection effectiveness is not concentrated in a narrow subset of conditions but instead remains broadly consistent across the operational envelope. Where minor heterogeneity appears (e.g., certain humidity sensor types under high dew-point proximity), feature-importance probes suggest interactions that can be mitigated by small rule-layer refinements or localized model recalibration. These observations reduce the risk of hidden failure pockets and support generalizability.

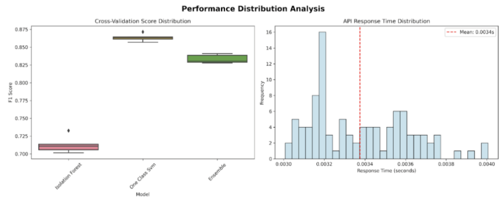

Fig. 24. Detection performance distribution across devices/locations/times

- **Implications for Deployment** Taken together, the results support deployment of **OC-SVM** as the default detector in tandem with the rule-based first layer. The system already meets latency, throughput, and aggregate detection targets but requires hardening against malformed inputs and better guardrails during data gaps to reach the robustness threshold for `UNCONDITIONAL_PASS`.

  Near-term priorities include:
  - Stricter schema enforcement at the gateway,
  - Explicit unit and range validators,
  - Conservative fallbacks for imputation,
  - Decision-time fusion that down-weights model scores when upstream rule violations indicate data quality risk.

  With these mitigations, the subsystem is well-positioned to progress from `READY_WITH_MONITORING` to fully `PRODUCTION_READY` in subsequent iterations.

  The anomaly detection system was tested on **September 2, 2025, at 09:58:38**, yielding a status of `CONDITIONAL_PASS` and an overall readiness level of `READY_WITH_MONITORING`. This result indicates that the system successfully meets the criteria for model performance and API reliability but still presents certain risks that require close monitoring.

  In particular, robustness against edge case scenarios remains a critical limitation, with a current success rate of only 42.9%, significantly below the target threshold of over 80%.

  Among the tested models, the **One-Class SVM** achieved the best performance, attaining an F1-score of 0.863 and a ROC-AUC of 0.943, thereby demonstrating strong capability in accurately detecting anomalies within the standard test dataset.
- **Security and Privacy** The system enforces end-to-end communication security via HTTPS (TLS/SSL), utilizes JSON Web Tokens (JWT) for authentication and access control, and stores data on MongoDB Atlas with encryption at rest and fine-grained authorization. During testing, no high-severity issues were observed that would compromise the confidentiality, integrity, or availability (CIA) of data. For field deployments, we have established additional hardening guidelines, including explicit token lifetimes and periodic secret rotation, per-user/per-device rate limiting, audit logging of security-relevant events, and a program of proactive security assessment (penetration testing) aligned with the OWASP Application Security Verification Standard (ASVS).

This chapter has presented a comprehensive analysis of the system's usability, model performance, and security posture. The usability evaluation demonstrated that the majority of core functions—including registration, device pairing, monitoring, notifications, statistics viewing, and data export—achieved high success rates, rapid task completion times, and strong user satisfaction. These findings confirm that the system provides an accessible and intuitive interface for both experienced and novice users. Areas requiring refinement, such as visual clarity of graphs, date-picker responsiveness, and enhanced multimodal feedback, were also identified to improve overall usability.

The anomaly detection module showed strong performance across multiple machine-learning models, with the One-Class SVM achieving the most balanced results (F1-score = 0.863; ROC-AUC = 0.943). While the subsystem met key performance indicators for detection accuracy and API reliability, edge-case robustness fell below the target threshold, highlighting the need for stricter data validation and improved error-handling mechanisms. Nevertheless, the results indicate that the system is operationally viable under a READY-WITH-MONITORING status and can be further strengthened through incremental improvements.

Additionally, API performance and scalability tests confirmed that the platform can support high-throughput, low-latency data processing suitable for real-time agricultural deployments. The security assessment revealed no high-severity vulnerabilities and validated the effectiveness of the implemented encryption, authentication, and access-control mechanisms, with recommendations for additional hardening in real-world environments.

In conclusion, the findings in this chapter demonstrate that the proposed IoT monitoring system exhibits strong potential for real-world deployment, offering reliable performance, robust security, and high usability. The insights gained from the evaluations provide a clear direction for future enhancements and support the system's progression toward full production readiness.

VI. CONCLUSION

This chapter synthesizes and interprets the findings of the study entitled "Advancing IoT-based Smart Farming through a Human-Centered Design Approach." It integrates functional results, user usability outcomes, and technical performance with the three research questions posed in Chapter 1: (1) the problems and constraints experienced by users—especially unstable sensor connectivity and difficulties in use; (2) design strategies to improve connectivity stability and UX/UI suitable for rural Thai contexts; and (3) the extent to which the improved system meets user needs in terms of stability, ease of use, and effectiveness in solving practical challenges. The conclusions presented here are grounded in evidence from prototype testing across eight usage scenarios, usability evaluation with six farmers in Chiang Rai Province, and measurements of system and anomaly-detection performance under varied loads and network conditions.

A. Summary of Key Findings across Three Evaluation Dimensions

This section synthesizes empirical evidence across three dimensions—Functional, Usability, and Technical Performance—drawing on eight scenario-based tests, usability sessions with six farmers in Chiang Rai, and benchmarking of the real-time pipeline and anomaly-detection module. Unless otherwise noted, functional results are reported as task success, time-on-task, and error counts; usability is assessed via SUS and qualitative observations; and technical metrics include mean and p95 latency, throughput, service error rate, model accuracy/precision/recall/F1/ROC-AUC, and edge-case pass rate. Key findings are summarized below.

- **Functional** The system supports end-to-end workflows—including onboarding, device addition and tracking, monitoring and alerting, historical/statistical analysis, and data export. An offline-first architecture with a buffer-and-sync mechanism maintained continuity despite high network latency or temporary disconnections.
- **Usability** Tests with target users indicate that a Thai-language interface, clear colour/icon status cues, and concise task flows reduced the learning burden. Initial stumbling blocks—discoverability of the export function and the salience of alert severity levels—were addressed by relocating the export button, adding an Export Preview, introducing quick filters and colour codes, and enlarging the touch targets of graph toggles. On retest, prior complaints did not recur; task completion times decreased and user confidence increased. The SUS scores reported in Chapter 5 corroborate this positive trend.
- **Technical performance**
  - Under standard load, mean end-to-end latency was approximately 0.0034 s, with $p95 \approx 0.0038$ s, zero service errors, and throughput of about 15,898.96 samples/second (batch size 100), indicating capacity to support dozens to low hundreds of devices concurrently.
  - For anomaly detection, the two-layer arrangement (rule-based thresholds and ML model) provided both interpretability for critical events and sensitivity to subtle signals.
  - The **One-Class SVM** performed best with:

$$\text{Accuracy} = 0.922, \quad \text{Precision} = 0.768,$$

$$\text{Recall} = 0.986, \quad \text{F1-score} = 0.863,$$

$$\text{ROC-AUC} = 0.943.$$

  - However, edge-case robustness remained below target, passing only 42.9% of designed cases, underscoring the need to strengthen input validation and stream-recovery policies.

B. Responses to the Research Questions and Goal Attainment

This section answers the three research questions (RQ1–RQ3) and assesses goal attainment against the predefined success criteria stated in Chapter 1 (e.g., real-time latency/p95 and zero service errors; usability via SUS and task metrics; anomaly-detection accuracy and edge-case pass rate). For each RQ, we summarize the supporting evidence from scenario-based functional tests, usability sessions with six farmers, and technical benchmarks of the real-time pipeline and anomaly-detection module.

- **RQ1—User problems** The study substantiated key barriers to adoption: connectivity instability and data loss/latency under adverse weather, insufficient discoverability of critical functions, and status/alert communication that was not yet inclusive for low digital-literacy users.
- **RQ2—Design solutions** The proposed offline-first architecture with buffer-and-sync, a two-layer anomaly-detection module, and a Thai, minimal UX/UI using colour/iconography and short task flows mitigated these gaps in practice.
- **RQ3—Effectiveness of the improved system** The improved system aligns well with user needs in responsiveness, proactive monitoring, and ease of use after UI/notification refinements. Robustness to edge cases has not yet met the$\geq$ 80% criterion, necessitating further work on data quality controls and prevention/recovery mechanisms.
- **Goal Attainment** The research objectives defined in Chapter 1 were evaluated against the empirical results reported in Chapter 5. The extent of goal attainment is summarized below:

Obj1 – Identifying user problems, This goal was fully achieved. The study confirmed key barriers to adoption, including unstable connectivity and data loss under adverse weather, difficulties in discovering critical functions such as data export, and insufficiently inclusive communication of status/alerts for users with low digital literacy.

Obj2 – Designing and developing an IoT architecture and UX/UI tailored to rural Thai farmers., This goal was fully achieved. The offline-first architecture with buffer-and-sync maintained continuity under unstable networks, while Thai-language interfaces with clear icons, colour cues, and concise task flows reduced the learning burden. UX refinements—such as relocating the export button, adding an Export Preview, introducing quick filters, and enlarging touch targets—resolved discoverability issues identified in early tests.

Obj3 – Evaluating technical performance and reliability of the system., This goal was partially achieved. The system demonstrated strong performance under standard conditions, with low latency (mean 0.0034 s, $p95 = 0.0038$ s), high throughput ($\sim 15,898$ samples/s), and no service errors.

The anomaly-detection module, led by **One-Class SVM**, achieved high detection quality:

$$\text{Accuracy} = 0.922, \quad \text{Recall} = 0.986,$$

$$F1 = 0.863, \quad \text{ROC-AUC} = 0.943.$$

Security mechanisms (`TLS/SSL`, `JWT`, encryption-at-rest) met ASVS guidelines. However, robustness under edge cases reached only 42.9%—well below the $\geq 80\%$ target—indicating the need for further hardening of input validation and recovery mechanisms.

Obj4 – Assessing whether the improved system meets user needs in practice., This goal was fully achieved. Across eight usage scenarios, farmers successfully completed tasks within required time thresholds, SUS scores confirmed usability improvements, and participants reported greater confidence and independence in use. The system provided proactive monitoring and responsiveness even under degraded network conditions, aligning with real-world user needs.

C. Contributions and Innovations

The study offers four principal contributions: (1) a design-and-development framework that fuses HCD with a real-time offline-first architecture tailored to rural Thailand; (2) UX/UI patterns for users with limited digital literacy (Thai language, colour codes, concise labels, clear status communication); (3) a two-layer anomaly-detection architecture that balances sensitivity and precision; and (4) an integrated evaluation framework (functional–usability–performance) that transparently maps research objectives to empirical evidence.

D. Practical Recommendations for Deployment

For field users, configure alert thresholds by crop/greenhouse and season; site devices to cover key micro-climates; enable multi-channel notifications (in-app + LINE/SMS) with a concrete incident playbook; and manage data in compliance with PDPA at farm/community level. For developers/operators, enforce schema validation at the edge, implement observability for latency/availability/throughput with explicit SLOs, prepare autoscaling and back-pressure for growth, and adopt proactive security measures (token lifetimes, key rotation, rate limiting, and periodic penetration testing aligned with ASVS).

E. Limitations and Threats to Validity

Limitations include the small user sample, partial reliance on synthetic/simulated sensors, limited diversity of crops/seasons/climates, and the lack of long-term field evaluation of solar power/battery ageing. The sub-target edge-case robustness poses a risk to alert reliability under certain malformed-data conditions, warranting caution in generalization.

F. Directions for Future Research

Future work should involve multi-season, multi-farm field trials with real hardware (e.g., LoRaWAN, NB-IoT, or Wi-Fi Mesh), alongside agronomic outcome metrics (yield, water/fertilizer efficiency, environmental impacts). Edge-case robustness should be raised to $\geq 80\%$ through stricter data contracts, robust parsing, self-healing and staged retry policies, and expanded synthetic/real test suites. Develop MLOps/Edge-AI pipelines for model/data drift detection and semi-automated retraining; make UX adaptive to user skills and device/lighting conditions; and assess farm-scale ROI and socio-environmental impacts to support wider adoption.

In summary, pairing a user-centered approach with a real-time architecture designed for intermittent networks can transform IoT into a practical, trustworthy tool for farmers—enhancing proactive monitoring, data-driven decision-making, and usability. While challenges remain in edge-case robustness and long-term field evaluation, the reported results establish a strong foundation for broader deployment and for academic and practical extensions that advance the long-term sustainability of Thailand's agricultural sector.

From the feedback from the respondents on the use of IoT systems in agriculture, it was found that there were both convenience and limitations. Some users stated that the system facilitated data management and saved resources such as water and energy. It also measured technical values such as EC and electricity accurately, which helped reduce costs in the long run. However, there were also problems mentioned, such as errors in device commands, difficulties in starting up the system, and energy problems, especially in cases of power outages or lack of necessary technical support. In addition, users provided suggestions on the need to develop a more stable system to reduce the impact of such problems. These comments reflected the importance of improving the IoT system to be ready for various uses and fully respond to farmers' needs. Improving the reliability and experience of using the IoT system revealed a variety of suggestions, such as a lack of knowledge about the application program. Some users wanted the software to be improved to have a security system and protect various data to create confidence in use. In addition, there was a demand to reduce the price or reduce the cost of using the system to an acceptable level that is suitable for agriculture in the area. There was also a proposal to establish a training center or teach about IoT sensors in the province to increase farmers' skills and understanding.

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APPENDIX SURVEY ITEMS

SECTION 1: DEMOGRAPHICS

Age: (Open-ended or select the appropriate answer)
Gender: (Male, Female, Other, Not specified)
Income level: What is the approximate income from farming per year?
Education level: (Secondary school, Vocational certificate, Bachelor's degree, Master's degree, Doctorate, Other)
Role in farming: (Farmer, System operator, Other)
Experience in farming: (Years of experience)
Farm area size: (Specify size)
Name of crops grown: (List crops)
IoT Usage: Are you currently using IoT or have you stopped?
Reasons for non-usage: Reasons for not using or stopping IoT.
IoT Experience Duration: How many years have you used IoT on your farm? (Less than 1 year / 1–3 years / More than 3 years)
IoT Devices: IoT devices or sensors currently in use.

SECTION 2: SYSTEM RELIABILITY

System Failure: How often do you experience system failures or sensor failures?
(Never / Rarely / Sometimes / Often / Very often)
Downtime: How many hours per week is the system down due to technical issues?
(0-1 hour / 1-5 hours / 5-10 hours / More than 10 hours)
Trustworthiness: How much trust do you have in the system for reliable data?
(Very little / Little / Moderate / Much / Very much)
Impact of Outages: How does the frequency of outages affect trust? (Open-ended)

SECTION 3: EASE OF USE

Ease of Data Interpretation: How easy is it to interpret IoT data?
(Very Easy / Easy / Moderate / Difficult / Very Hard)
Usefulness: How useful is the IoT system in farming?
(Very Little / Little / Moderate / Much / Very Much)
Training and Support: How much training do you need?
(None / Little / Moderate / Very Much)

SECTION 4: DATA ACCURACY

Field Reflection: How accurate is the system in field conditions?
(Always correct / Almost always correct / Sometimes correct / Rarely correct / Never correct)
Distortion: How often do system recommendations mismatch results?
(Always / Often / Sometimes / Rarely / Never)

SECTION 5: ENVIRONMENTAL RESPONSE

Environmental Impact: Factors affecting system performance:
(Storm / Power Outage / High Temperature / Heavy Rain / Dust / Other)
Severity: How much do environmental factors affect performance?
(Not at all / A little / Moderate / A lot / Very much)
Adaptability: How well does the system adapt to weather changes?
(Very good / Good / Average / Poor / Very poor)
Power Stability: How stable is the power supply for IoT?
(Very unstable / Unstable / Moderate / Stable / Very stable)
Weather Accuracy: How much does weather impact sensor accuracy?
(Very little / Slightly / Moderate / Much / Very much)
Physical Disturbances: How often do disturbances affect system reliability?
(Not at all / Slightly / Moderately / Much / Very much)
Decision Making: How do environmental factors affect decisions? (Open-ended)

SECTION 6: MAINTENANCE AND SUPPORT

Maintenance Frequency: How often do you perform maintenance?
(Weekly / Monthly / Quarterly / Annually)
Troubleshooting Ease: How easy is troubleshooting IoT issues?
(Very Easy / Easy / Moderate / Difficult / Very Hard)
Recalibration Needs: How often does the system need recalibration?
(Weekly / Monthly / Quarterly / Annually / Never)

SECTION 7: COST AND BENEFITS

Productivity Impact: How does the system affect productivity?
(Increased significantly / Increased slightly / No change / Decreased slightly / Decreased significantly)
Cost Impact: How does the system affect costs?
(Decreased significantly / Decreased slightly / No change / Increased slightly / Increased significantly)
Expectations: Are benefits in line with expectations?
(Exceeds expectations / Consistent / Below expectations)

SECTION 8: OPEN-ENDED SUGGESTIONS

Overall Experience: Describe your experience using IoT. (Paragraph)
Improvement Suggestions: Suggestions to improve reliability. (Paragraph)